

Working with Graphics

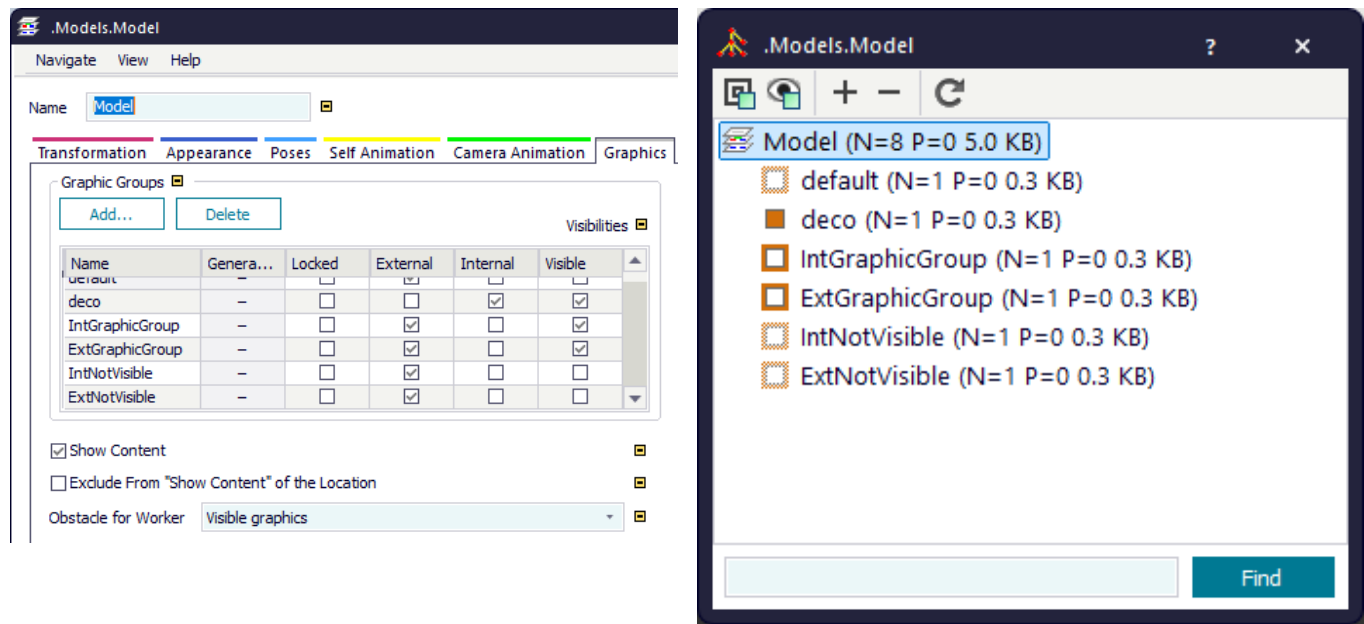
Work with Graphics

Plant Simulation displays objects with graphics, which are structured in one or several graphic groups.

Graphic groups of objects and *Frames* consist of one or more graphics and have a name that is unique for this object. This is similar to the 2D icons of the simulation objects. You can define a graphic group to either be external or internal.

External graphic groups represent the object toward the outside, while **internal graphic groups** are used to decorate the inside of the object. The properties which graphic groups are internal or locked are, together with the graphic structure, an object property that can be inherited. A graphic group can be permanently shown or hidden to allow switching between alternative visual representations. Each simulation object or animatable object contains at least one external graphic group named **default** and optionally any number of alternative or additional graphic groups, which you can show or hide independent of each other.

In addition, a *Frame* initially contains one internal graphic group named **deco**. The visibility of the graphic groups is an object property which can be inherited independent from the graphic structure and the properties **internal** and **locked**. You can create, delete, or configure graphic groups on the tab **Graphics** [tab] in the dialog **Edit 3D Properties** and in the dialog **Show Graphic Structure** [button].



Icon looks like this	means that the graphic group is
☐ default	visible
☒ deco	visible

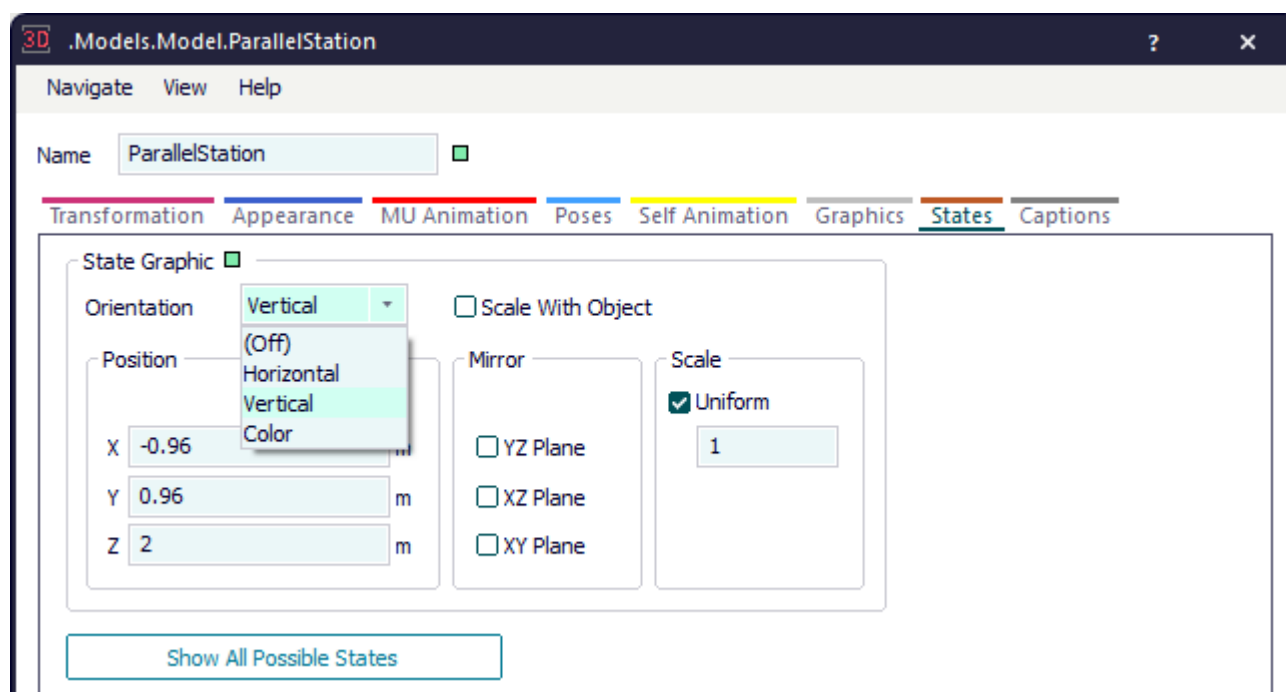
Icon looks like this	means that the graphic group is
 IntNotVisible	not visible, as selected
 ExtNotVisible	not visible, as selected

You can delete and edit each graphic of a graphic group individually. To modify a graphic you can either change it with the mouse in a 3D window or you can specify the new **position**, the **rotation**, the **scaling**, or the **material** in the dialog **Edit 3D Properties** of the selected graphic.

Note

If you change any graphic or the graphic structure of a class object, this also changes all inherited objects. If you change the graphics of a derived object, this only changes the object in which you make the changes because 3D deactivates graphic inheritance if it has not been deactivated already. As deactivating graphic inheritance causes graphics to be duplicated, we strongly advise you to make changes in the class object as far as possible.

You can create, replace or delete the state graphic on the tab **States** in the dialog **Edit 3D Properties** by selecting the orientation of the states for the state graphic: **Horizontal**, **Vertical**, or **Off**. When you change this setting, 3D always deletes all state graphics and creates new state graphics afterward, which are arranged according to your selection.



Back to [Visualizing the Material Flow](#)

Inherit Graphics

Object graphics are, as other object properties, data of the object that can be inherited. This means that objects, which inherit their graphics, share the graphics with their parent object and thus preserve memory. Hence graphic changes in the class objects are automatically also visible in derived objects whose graphic inheritance is activated.

Graphic changes in derived objects deactivate graphic inheritance. Then the derived object contains its own, now changed, copy of the object graphic of the parent object and requires additional memory.

The following graphic changes deactivate graphic inheritance:

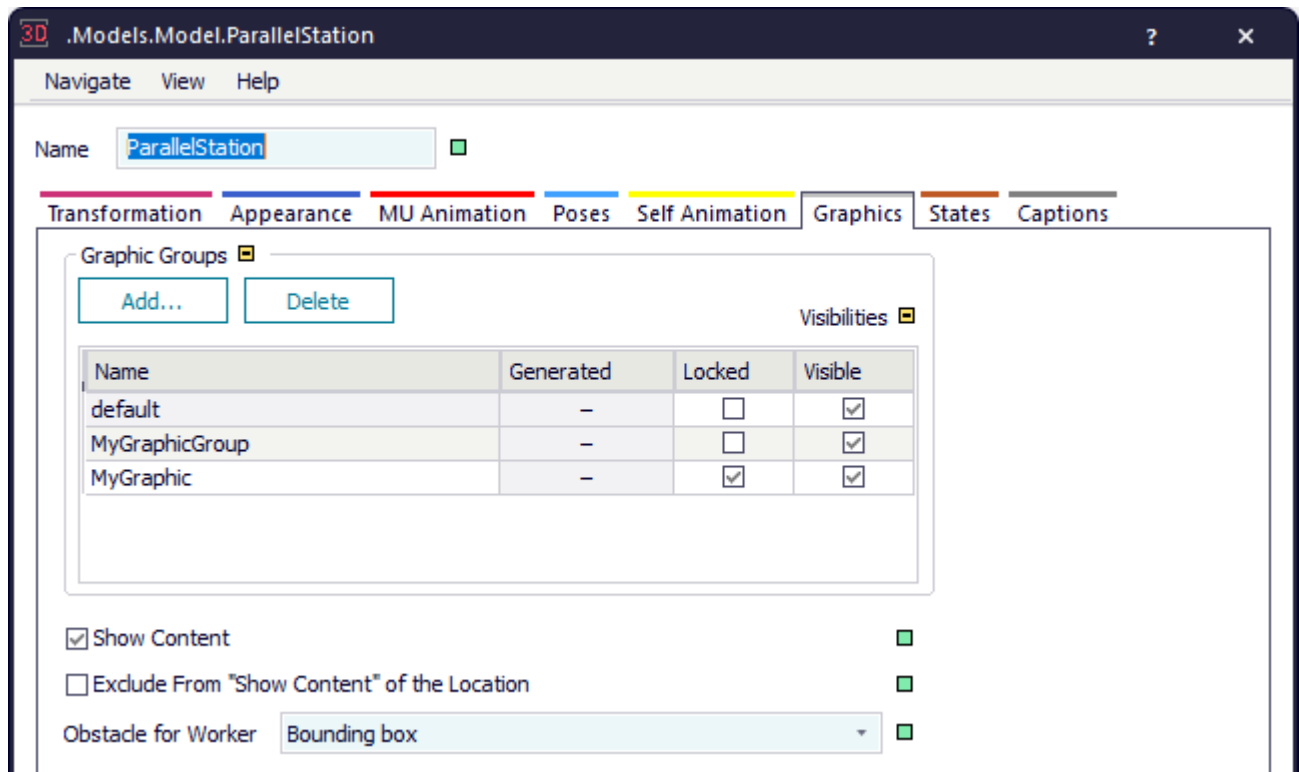
- Adding or deleting graphic groups or graphics.
- Transforming graphics, i. e., moving, rotating, zooming in or out of graphics.
- Changing the material of graphics.
- Grouping or ungrouping of a graphic group.
- Renaming of a graphic group.
- Changing the graphic group settings **internal** or **locked**.

Note

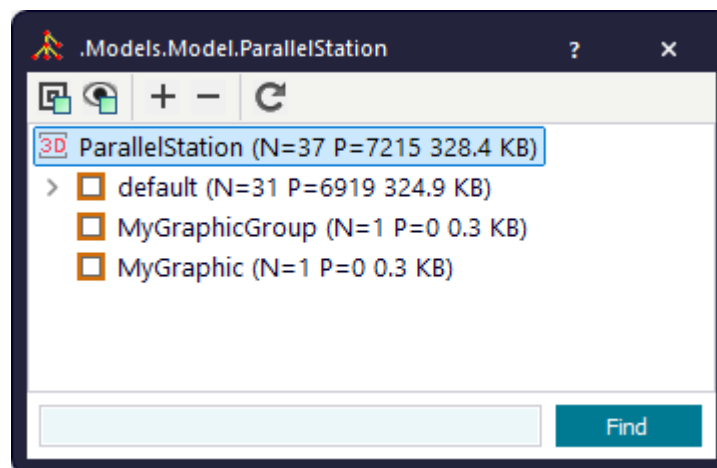
As turning graphic inheritance off makes your model file larger and slower from a certain point on, we strongly advise you to **make graphic changes in the class object** as far as possible.

You can activate or deactivate graphic inheritance as follows:

- By clicking the inheritance check box under **Home > Edit 3D Properties > Graphics > Graphic Groups** so that it looks like this .



- By clicking **Inherit Graphic Groups**  in the dialog **Show Graphic Structure**.



When you turn graphic inheritance on, the object will use the graphics of its parent again and cancel your changes.

When you leave graphic inheritance off, the object keeps its changed graphics of the instance and thus uses more memory.

See also

[_3D.InheritGraphics \[SimTalk\]](#)

Using Inheritance

Back to [Work with Graphics](#)

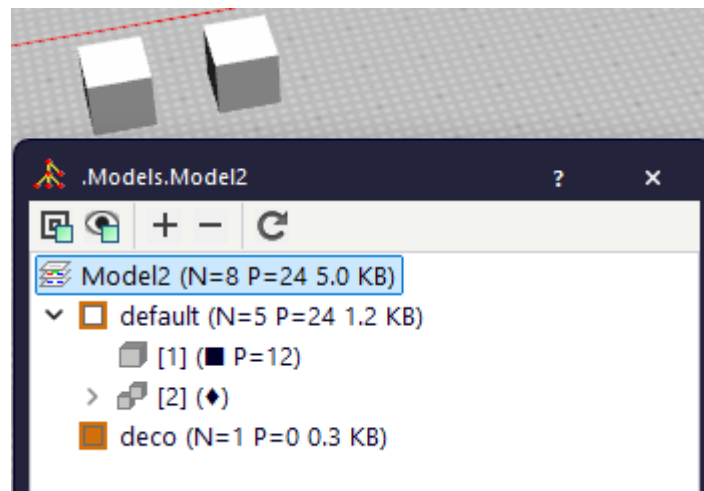
Duplicate Graphics

You can duplicate the graphic you selected in the *3D window*.

To duplicate the selected graphic, hold down **Ctrl+Shift** and click the left mouse button in the *3D window* at a different position.

Plant Simulation does not create a copy of the graphic, but the new graphic references the duplicated graphic. This considerably reduces memory consumption.

Plant Simulation shows that the graphic is referenced with a diamond  in the dialog [Show Graphic Structure](#).



Edit Grouped Graphics

You can edit grouped graphics, i.e., the graphic group as a whole.

To move, rotate, or scale several graphics at the same time while retaining their positions to each other, you can group them and manipulate the group as a whole.

Note

If you change any graphic or the graphic structure of a class object, this also changes all inherited objects. If you change the graphics of a derived object, this only changes the object in which you make the changes because *Plant Simulation* turns off graphic inheritance if it has not been deactivated already. As deactivating graphic inheritance causes graphics to be duplicated, we strongly advise you to make changes in the class object as far as possible.

To group several graphics:

- Hold down **Shift** and click the objects to select them or drag a marquee over them.
- Click  **Group Graphics** on the **Edit** ribbon tab or press **Ctrl+G**. *Plant Simulation* creates a single new group.
- Then, move, rotate, scale, or zoom the group as you do a single graphic.
- Naturally you can also move, rotate, scale, or zoom the group on the **Tab Transformation** in the dialog **Edit 3D Properties**.

To restore the previous structure of the graphics:

- Select the newly created group.
- Click  **Ungroup Group of Graphics** on the **Edit** ribbon tab or press **Ctrl+U** to replace this group with the contained graphics again.

Compare [Move an Object Precisely](#)

Compare [Move an Object With Keyboard Keys](#)

Compare [Move an Object With the Mouse](#)

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Set the Material of a Graphic

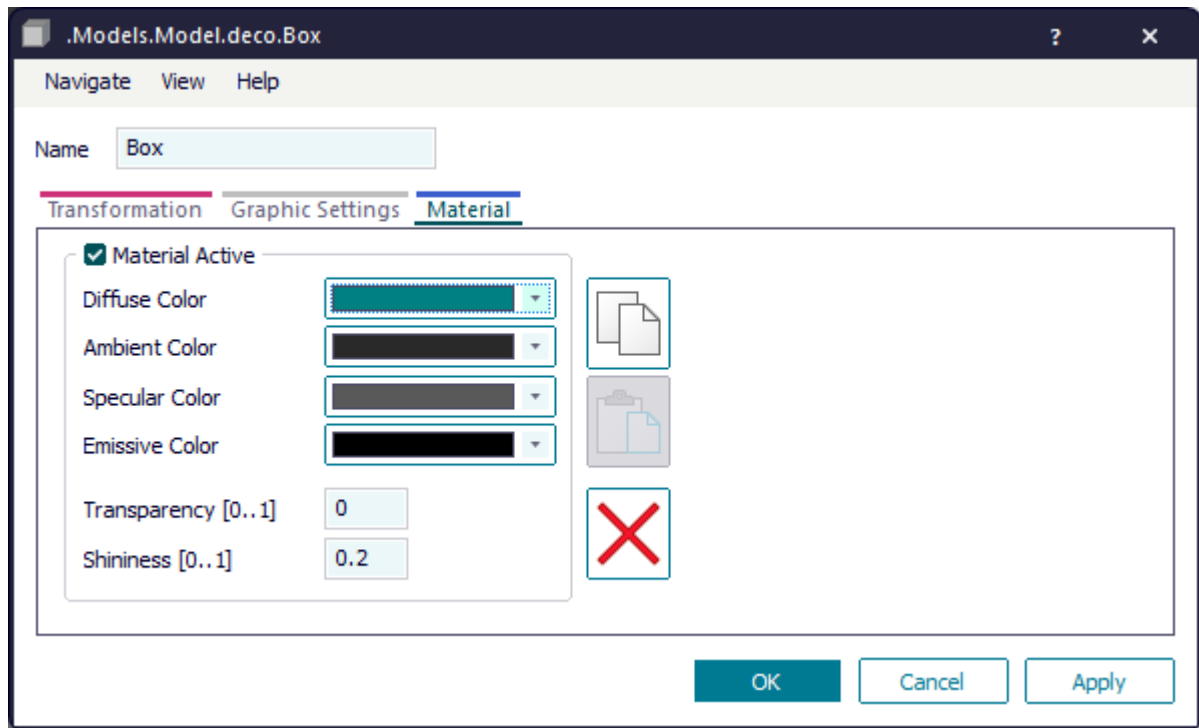
You can define the surface qualities, the appearance, i. e., the material of the graphic: How well it reflects light, what color it reflects, and what color it emits.

Note

If you change any graphic or the graphic structure of a class object, this also changes all inherited objects. If you change the graphics of a derived object, this only changes the object in which you make the changes because *Plant Simulation* deactivates off graphic inheritance if it has not been deactivated already. As deactivating graphic inheritance causes graphics to be duplicated, we strongly advise you to make changes in the class object as far as possible.

To define properties of the material of a graphic:

- Select the graphic and click  **3D Properties** on the **Home** ribbon tab.
- Click the **Tab Material** and edit the material of the graphic.



- To assign a material to the graphic, select **Material Active**.
- To apply a **Diffuse Color**, click the down arrow and select a color in the color selector.

The **diffuse color** is the reflection of light from an uneven surface so that an incident ray is seemingly reflected at a number of angles. It is the complement to specular reflection. If a surface is completely nonspecular, the reflected light will be evenly spread over the hemisphere surrounding the surface.

- To apply an **Ambient Color**, click the down arrow and select a color in the color selector.

The **ambient color** is the color of the light reflected from the object when it is lit by another ambient object in the scene.

- To apply a **Specular Color**, click the down arrow and select a color in the color selector.

The **specular color** is the reflected color of the object's highlights.

- To apply an **Emissive Color**, click the down arrow and select a color in the color selector.

The **emissive color** is the color emitted by the object. A lamp shade might have a base color of yellow. When you turn the lamp on, though, the emissive color might be white.

- To apply a **Transparency**, enter a value between 0.0, for opaque, and 1.0, for completely transparent, into the text box.

The **transparency** sets how opaque or clear the graphic is. Water, for example, might be more clear than opaque.

Note

If your model contains several partially or fully transparent graphics one directly behind the other from your current view point, some or all of these graphics might look wrong or might seem to be missing.

This is due to heuristics without which the rendering performance would drastically decrease.

- To apply a **Shininess**, enter a value between 0.0, for a very dull surface, and 1.0, for a highly polished surface, into the text box.

The **shininess** sets how sharp light is reflected by the object. A billiard ball, for example, has a large shininess, making for small and sharp highlights.

- To apply the material you selected to the graphic, click **Apply**.

You also can:

- **Copy the Current Material Settings** by clicking .

- **Paste a Previously Copied Material** by clicking .

- **Remove the Materials in Nested Graphics** by clicking .

Note

When you define materials for a graphic in the graphic structure at different levels in the structure, *Plant Simulation* always uses the material that is located **furthest down** in the structure for the actual rendering.

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Use a Different Graphic for an Object

In some cases it will suffice to model the machines of your plant with the default graphics we provide for the material flow objects and for the mobile objects. In other cases you will want to replace these default graphics with pictures that resemble the actual machines used in your plant. In some cases you might even have the actual graphics files of these machines from the builder of the machine, which you can then use.

Note

If you change any graphic or the graphic structure of a class object, this also changes all inherited objects. If you change the graphics of a derived object, this only changes the object in which you make the changes because *Plant Simulation* deactivates off graphic inheritance if it has not been deactivated already. As deactivating graphic inheritance causes graphics to be duplicated, we strongly advise you to make changes in the class object as far as possible.

We saved the graphics that are part of your program package in *s3d* format, we already defined animation paths for most of them, and we scaled the objects correctly, so that they match the size of the other objects.

Instead, you can also select a graphic which you created in 3D with the command **Export Scene > Export Object**.

To show an object with a different graphic, you can do one of the following:

- **Exchange the object graphic with another pre-defined object graphic**

Select the object, and click  **Exchange Graphics** on the **Edit** ribbon tab of the *3D window* or select the command on the context menu.

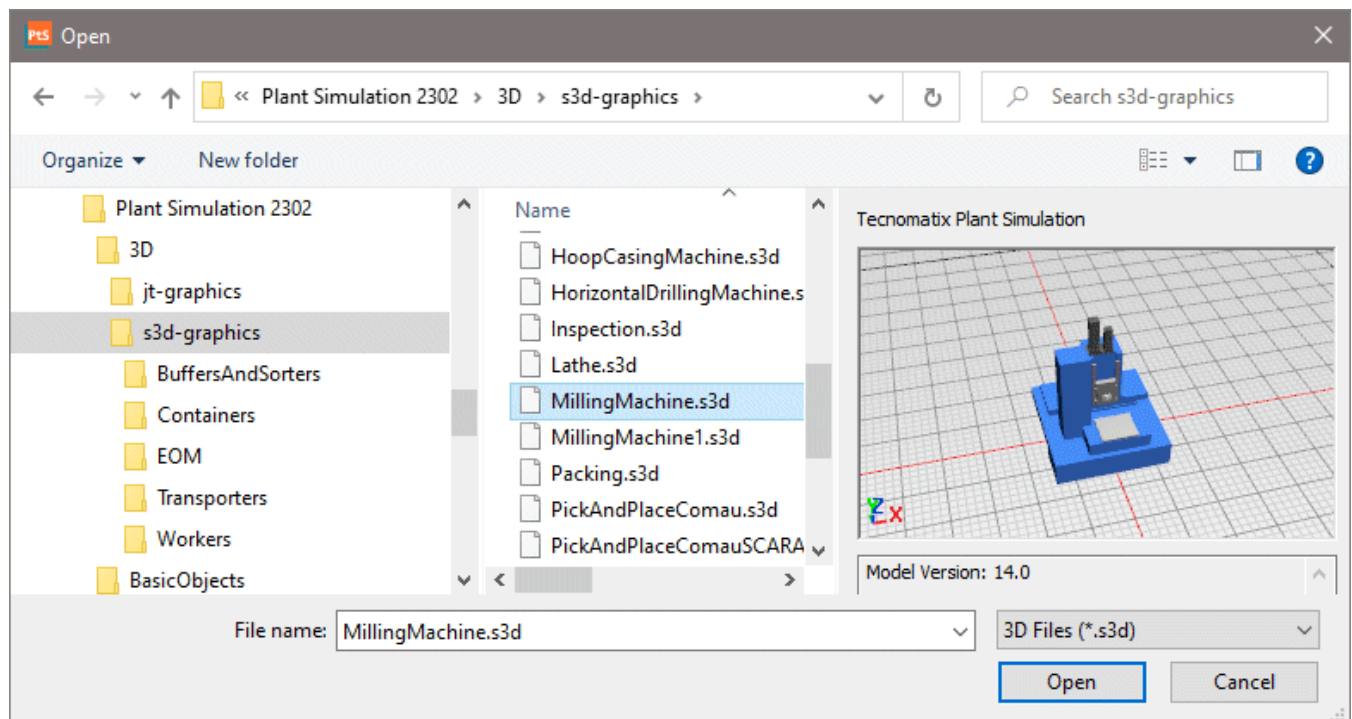
Plant Simulation opens the dialog **Exchange Graphics**. By default it opens to the folder where you installed *Plant Simulation*, namely **C:\Program Files\Siemens\Tecnomatix Plant Simulation XX\3D\s3d-graphics**. Change to the folder **Plant Simulation\3D\s3d-graphics** and select one of the *s3d* files. You might have to open one of the folders within that folder.

Click **Open** to replace all graphic and animation data of the selected object with data from the selected *.s3d*-file. This means that all graphic groups, the state graphic, and the animation attributes will be removed from the selected object and graphics and animation attributes of the saved object will be added instead.

Select the *.s3D* graphic that you want to use.

Note

Make sure that **Preview**  is selected, so that you can see what the graphic looks like.



To exchange graphics, you can also drag an *.s3d*-file from a **File Explorer** window and drop it onto a *3D window*.

- **Insert the graphic of an existing graphics file**

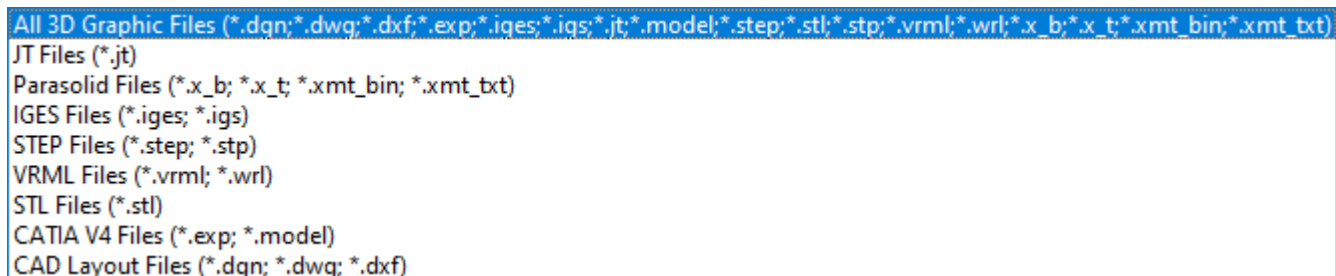


Select the graphic, and click **Graphics** on the **Edit** ribbon tab of **3D**.

Plant Simulation opens the dialog **Open**. By default it opens to the folder where you installed *Plant Simulation*, namely **C:\Program Files\Siemens\Tecnomatix Plant Simulation XX\3D\jt-graphics**.

Select the *.jt* graphic that you want to use.

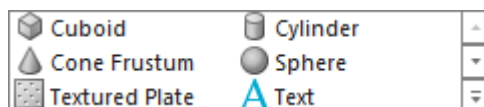
Instead, you can also import a graphic which you created in **3D** with the command **Export Scene** > **Export Graphics**. Or you can import a graphics file that you exported from another 3D modeling program in a format that *Plant Simulation* can import.



After *Plant Simulation* inserted the imported graphics, it will add them to the selected graphic group while keeping any already existing graphics. Delete any graphics that you do not need any longer.

To import graphics you can also drag a file of a supported graphic format from a **File Explorer** window and drop it onto a **3D window**.

- **Create your own object graphic**



Open the object, click **Insert Shape** on the **Edit** ribbon tab of **3D** and select the shape that you want to create.

After *Plant Simulation* inserted the shape you define, it will add it to the selected graphic group while keeping any already existing graphics. Delete any graphics that you do not need any longer.

You can also use graphics from other graphic groups by opening the dialog **Show Graphic Structure** [context menu] of the source object, copying the desired graphic with the respective context menu command, and then inserting it into the destination object with **Ctrl+V**.

- **Create a complex graphic** by inserting multiple shapes or imported graphics and position, rotate, scale, or color them.

If you are working with a simulation model, which you created in a previous version, you can use the icons of the **2D** objects and map them to a **Textured Plate** in **3D**.

Note

After modifying the object graphics by importing, creating, or transforming them, you might have to adjust the animation paths or the animation points of the 3D object to the new overall graphic. You do not have to do this if you used the command [Exchange Graphics](#).

Compare [Import JT Graphics Representing an Object](#)

Back to [Work with Graphics](#)

[Video on YouTube](#)

https://youtu.be/K-NEPvShU10?si=ChF_2xnoMuE1Fq7q&t=730

Creating Your Own Objects

Create Your Own Objects

You can create your own graphics to meet your modeling needs in your simulation mode.

Proceed as follows:

- **Duplicate** or **derive** any of the class objects that we provide in [Class Library](#) and that have similar properties as the one you would like to create.

You can then edit the 3D properties of this new class object, so that it fits your needs, for example by [using another graphic](#) ([installation folder > 3D > s3d graphics](#)), [work with animation paths](#), etc.

- **Import a three-dimensional graphic** from the *Plant Simulation* graphics library that is part of your program package ([installation folder > 3D > jt graphics](#)), from your own graphics library, or from a graphics program that can save files in a format that *Plant Simulation* can [import](#), into your simulation model. After you imported a graphic that you downloaded from the internet or created in a 3D modeling program, you might have to scale, rotate and move it to the origin of the scene.

You can then transform the graphic into a simulation object. To do so, click the graphic with the right mouse button and select [Make Simulation Object](#). If the thus created simulation object can contain additional simulation objects, you might also want to adjust its animation paths to the changed object graphic, compare [Work with Animation Paths](#).

- **Model your own 3D graphic** by [inserting a shape](#) as often as required, position, rotate, scale or color it. You can also use any graphics from other graphic groups and edit them. If you want to use several graphics for your object graphic, you have to [group](#) these individual graphics to be able to transform the group into a simulation object later on. Click the graphic with the right mouse button and select [Make Simulation Object](#). In case the thus created simulation object can include additional simulation objects, you should also adjust its simulation paths to the changed object graphic, compare [Working with Animation Paths](#).
- **Create a simulation object with animation** to make your simulation model look as close as possible as it is in reality.

Note

If you change any graphic or the graphic structure of a class object, this also changes all inherited objects. If you change the graphics of a derived object, this only changes the object in which you make the changes because *Plant Simulation* deactivates off graphic inheritance if it has not been deactivated already. As deactivating graphic inheritance causes graphics to be duplicated, we strongly advise you to make changes in the class object as far as possible.

Compare [Inherit Graphics](#)**Back to [Work with Objects](#)****Import a 3D Graphic**

You can import graphics into your 3D simulation model to create your own 3D objects.

3D can import these file formats:

```
All 3D Graphic Files (*.dgn;*.dwg;*.dxf;*.exp;*.iges;*.igs;*.jt;*.model;*.step;*.stl;*.stp;*.vrm|;*.wrl;*.x_b;*.x_t;*.xmt_bin;*.xmt_txt)
JT Files (*.jt)
Parasolid Files (*.x_b; *.x_t; *.xmt_bin; *.xmt_txt)
IGES Files (*.iges; *.igs)
STEP Files (*.step; *.stp)
VRML Files (*.vrm|; *.wrl)
STL Files (*.stl)
CATIA V4 Files (*.exp; *.model)
CAD Layout Files (*.dgn; *.dwg; *.dxf)
```

- All 3D Graphic Files
- JT Files (*.jt)

Note

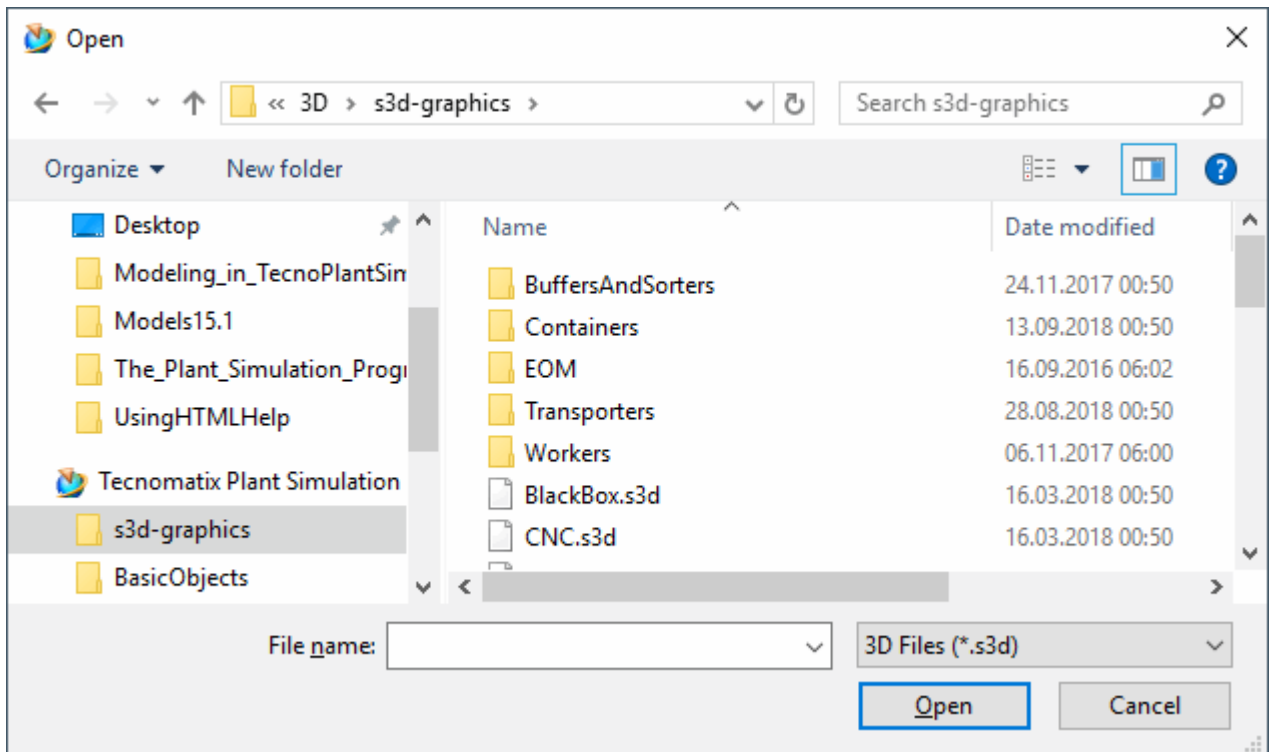
We recommend to use the JT format whenever possible.

- Parasolid Files (*.x_b; *.x_t; *.xmt_bin; *.xmt_txt)
- IGES Files (*.igs; *.iges)
- STEP Files (*.step; *.stp)
- VRML Files (*.wrl)
- STL Files (*.stl)
- Catia V4 Files (*.exp; *.model)
- CAD Layout Files (*.dgn; *.dwg; *.dxf)

To import a **graphic that contains animation structures** which you, or a colleague, defined, you can, for example, import **JT Files**:

- Click **Import Graphics** on the **Edit** ribbon tab. Select the respective file type, navigate to the folder that contains the graphic, select it and click **Open**.

These graphic files can either be files that are part of your program package or files you yourself created with the command **Export Graphics**.



- To replace the graphic of an object with another graphic that contains animation structures:
Click the object with the right mouse button and select **Exchange Graphic**. Select **3D Files**, navigate to the folder that contains the graphic, select it and click **Open**.

To import a **VRML File**.

Click **Import Graphics** on the **Edit** ribbon tab. Select **VRML Files (*.vrm|; *.wrl)**, navigate to the folder that contains the graphic, select it, and click **Open**.

Compare **Add a Background Graphic to the Frame**

Back to **Create Your Own Objects**

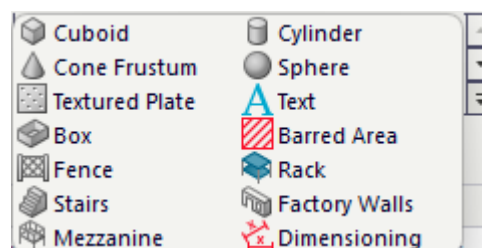
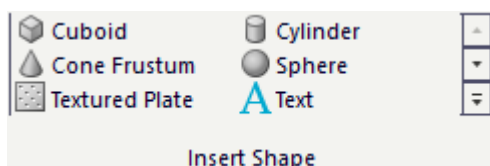
Video on YouTube

https://youtu.be/h2cFAH46ESQ?si=eKZLbWqgeP17LH_F&t=125

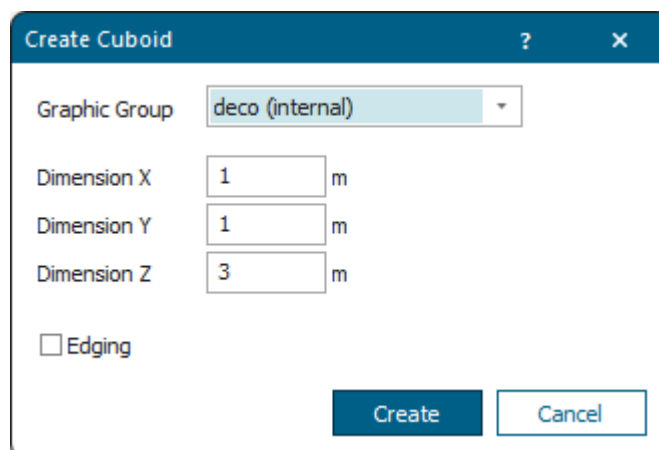
Model Your Own 3D Graphic

You can model your own 3D graphic for a simulation object, by inserting one or several of these **Shapes**.

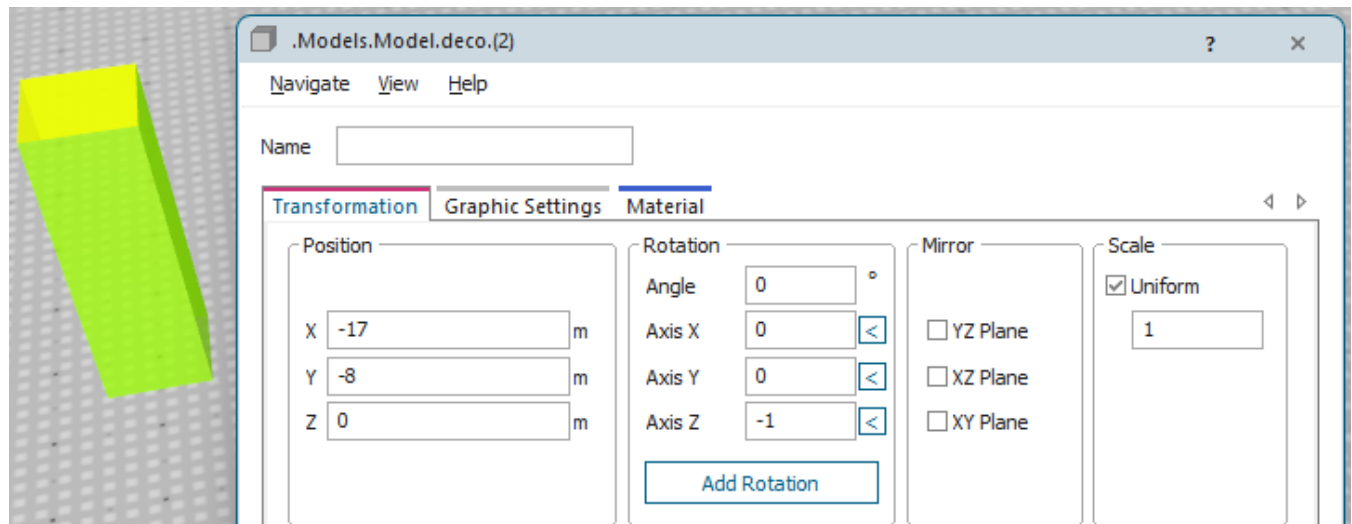
Barred Area, Box, Cone, Cuboid, Cylinder, Dimensioning, Factory Walls, Fence, Mezzanine, Rack, Sphere, Stairs, Text, or Textured Plate.



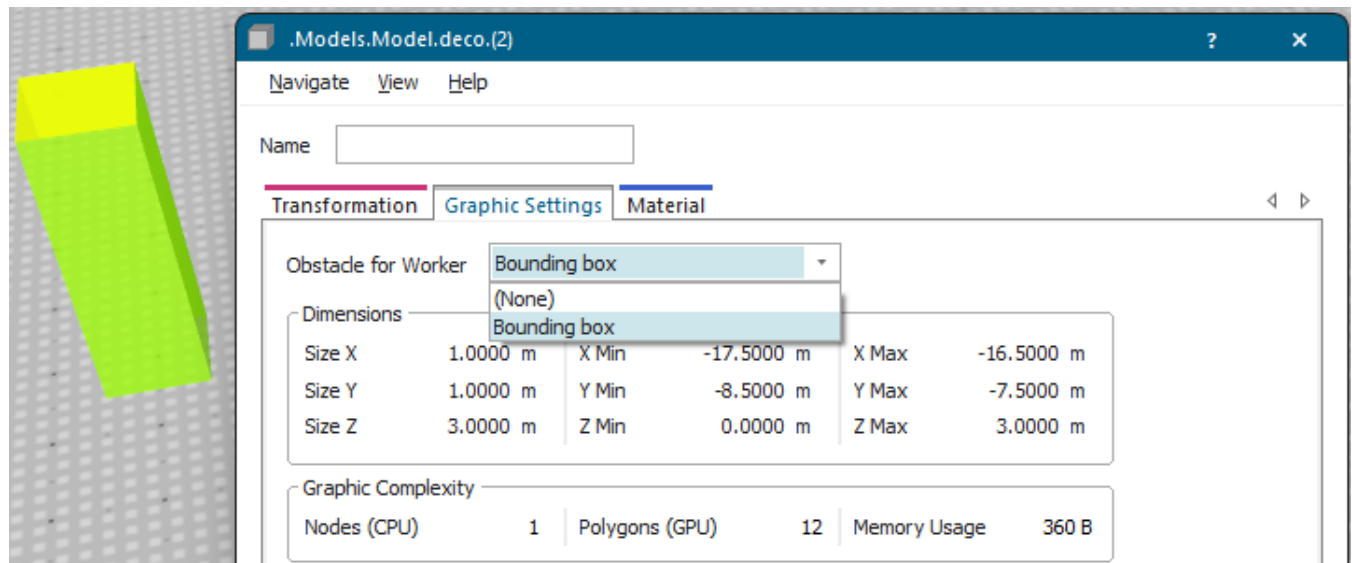
- Click the shape you want to create. In our example we create a cuboid .
- Select the **Graphic Group** to which you want to add the cuboid.
- Type in the **Dimension X/Width**, the **Dimension Y/Depth**, and the **Dimension Z/Height** of the cuboid.



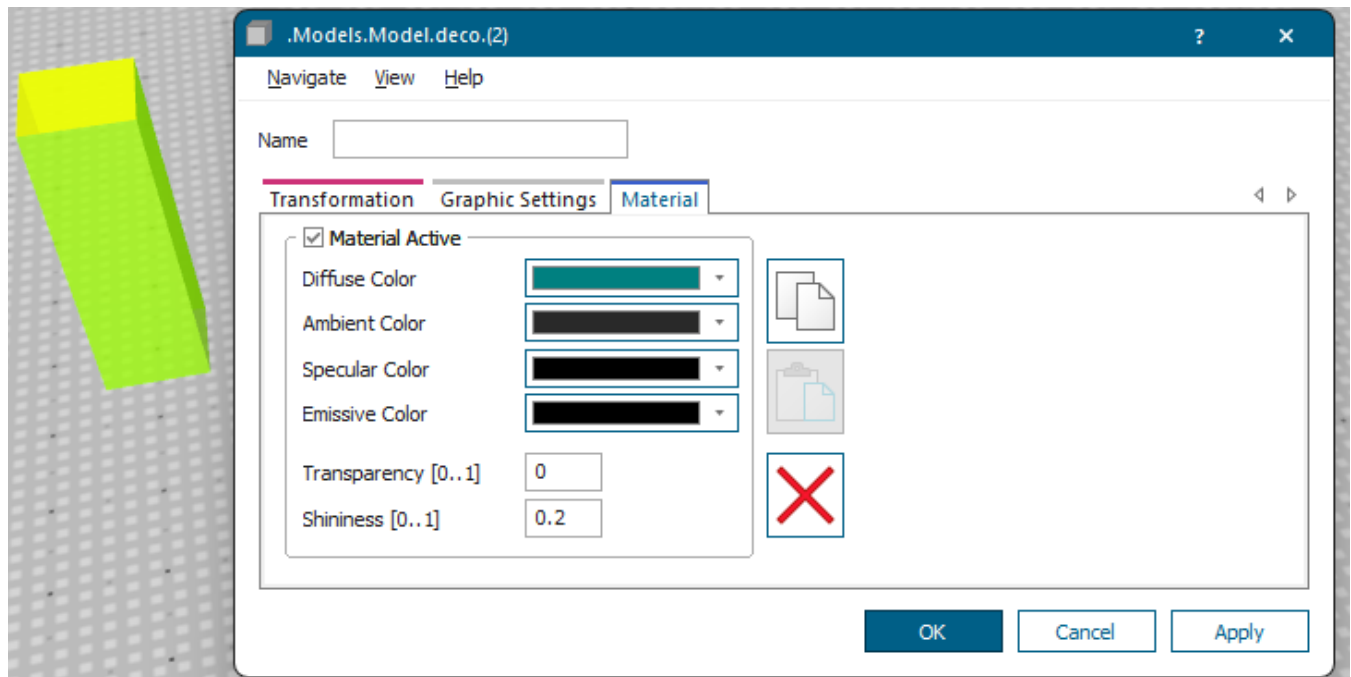
- To create the cuboid, click **Create**. 3D attaches the cuboid to the mouse pointer. Drag the mouse to the position at which you would like to insert it and click the left mouse button. To cancel inserting, click the right mouse button or press **Esc**.
- To change the **position**, the **rotation**, or the **scale** of the shape after you inserted it, select it and click **3D Properties > Tab Transformation** on the **Home** ribbon tab. Then change the settings according to your needs.



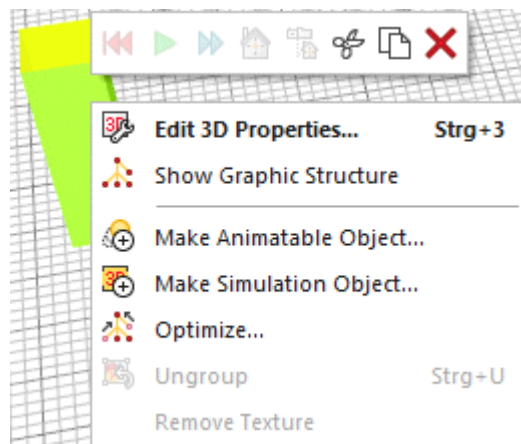
- To set if the cuboid is an **obstacle for the worker**, select it. Then, click **Edit 3D Properties** on the **Home** ribbon tab and click the **Tab Graphic Settings**.



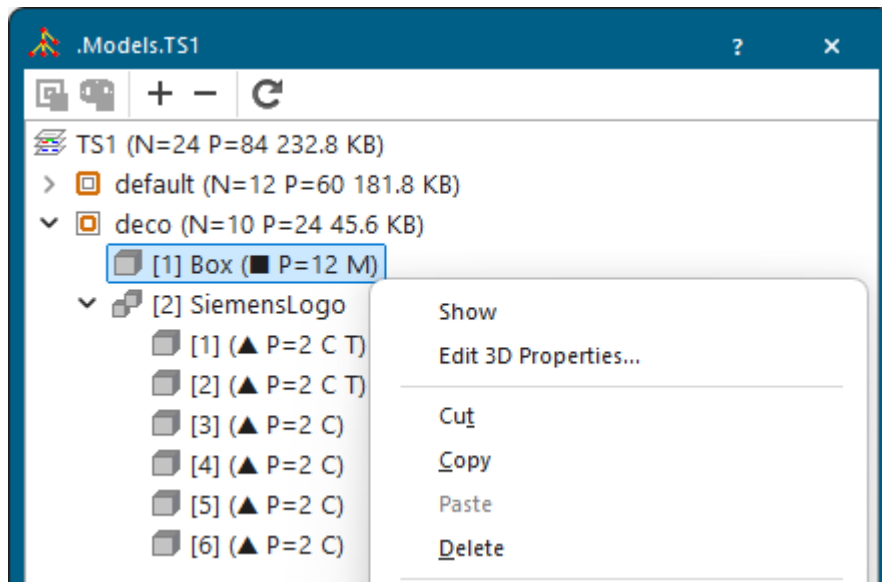
- To assign a color and a material, select the cuboid and click **3D Properties** on the **Home** ribbon tab. Then, click the **Tab Material**.



- To create a more complex graphic, insert several shapes and position, rotate, scale or color them. In addition to the pre-defined shapes you can copy graphics from other graphic groups or import graphics from graphic files.
- To delete the shape, click it with the right mouse button and click  on the mini toolbar or select **Show Graphic Structure**.



If the graphic group to which you added the shape, **deco** in our case, is not expanded already, expand it. Click the node of the shape you want to delete with the right mouse button. Select **Delete**.



Note

When you start to insert a newly created shape into an external graphic group, *Plant Simulation* automatically activates **Show External Graphic Groups**  until you click the destination position so that you can align the new graphic, which is attached to the mouse pointer, with all other graphics of the destination graphic group.

Compare [Create a Textured Plate](#)

Compare [Create a Non-textured Plate \[example\]](#)

Compare [Model a Bottle](#)

Back to [Create Your Own Objects](#)

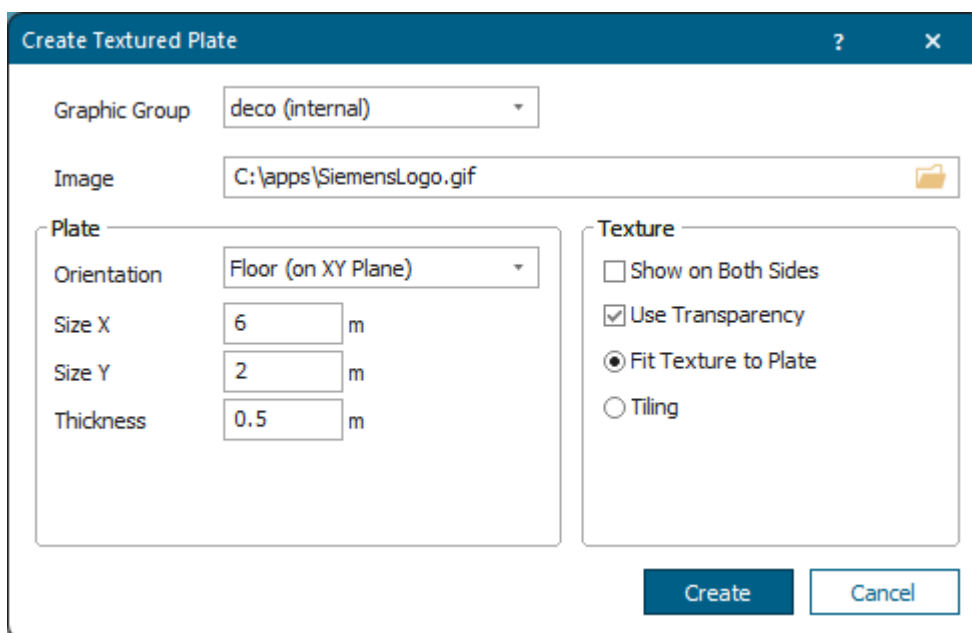
Create a Textured Plate

You can create a textured plate in your simulation model.

Proceed as follows:

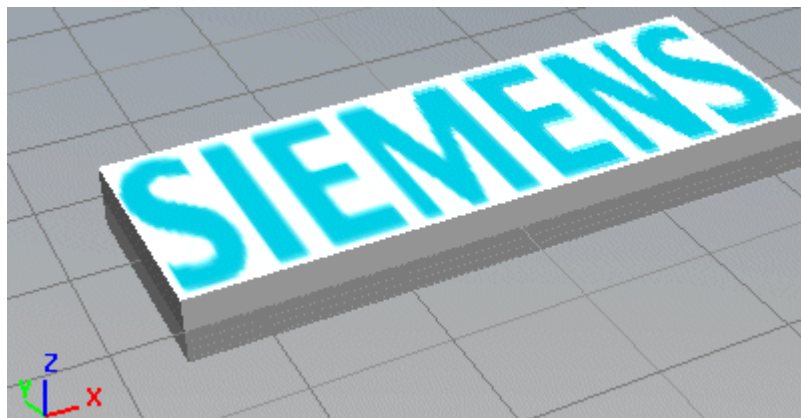
- Click  **Textured Plate** on the **Edit** ribbon tab of the *3D window*.

For our example we use the rectangular image with the **SIEMENS** logo. We selected these settings:

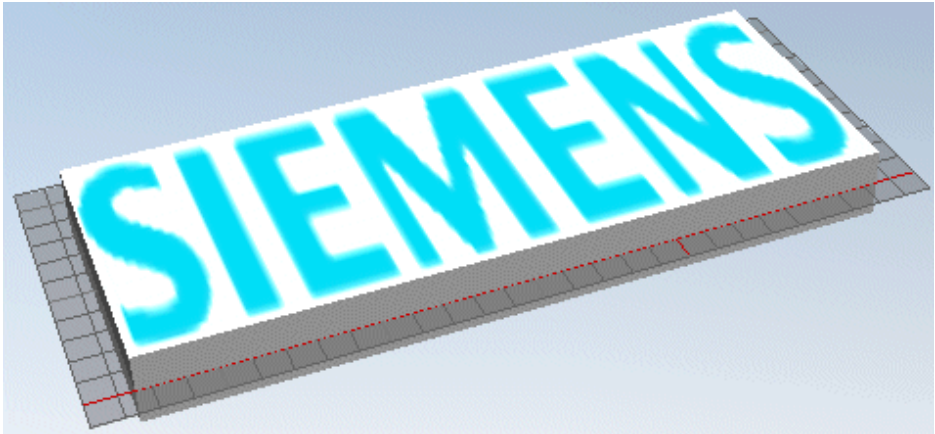
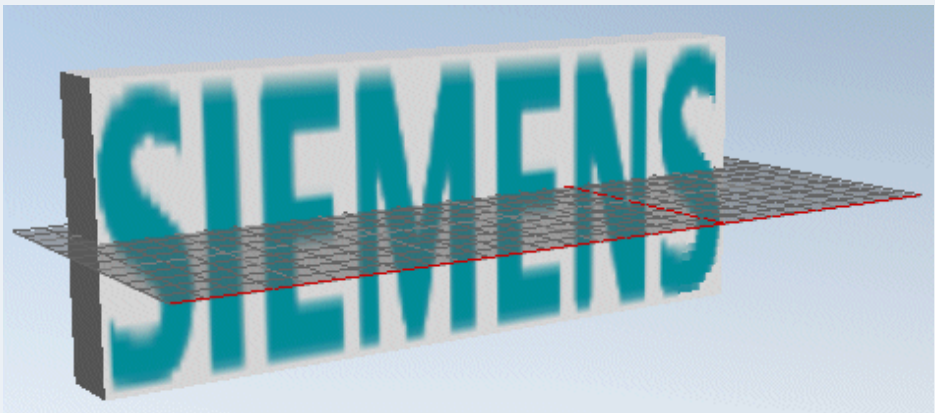
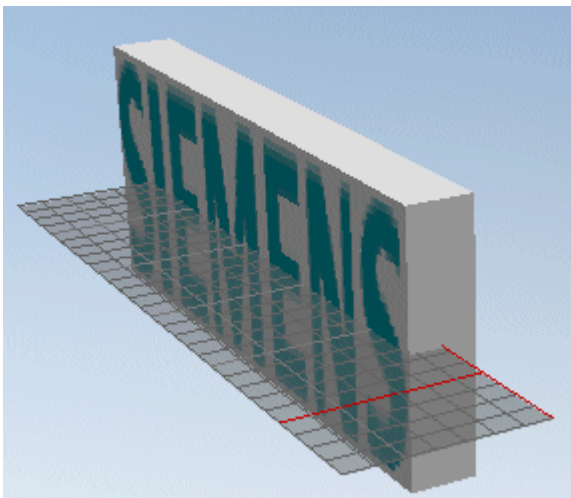


- Click **Create**. 3D attaches the textured plate to the mouse pointer. Drag the mouse to the position at which you would like to insert it and click the left mouse button. To cancel inserting, click the right mouse button or press **Esc**.

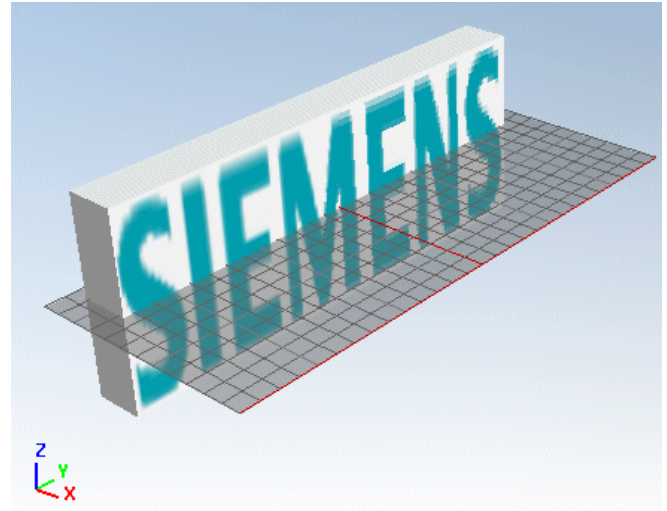
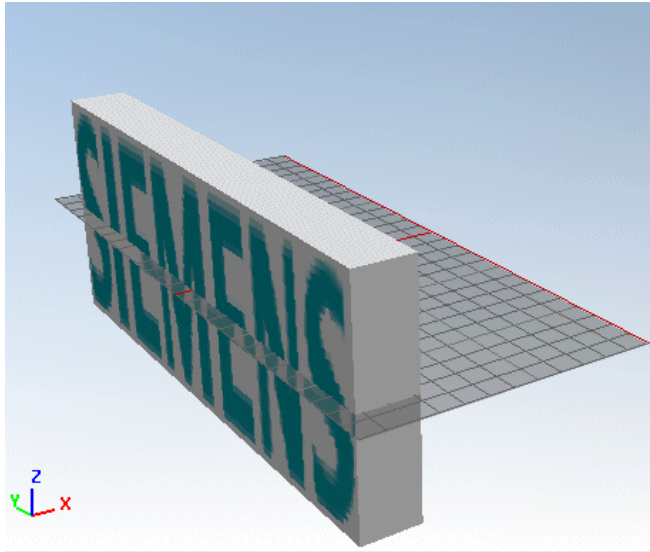
The inserted plate looks like this with the settings **Orientation > Floor** and **Fit Texture to Plate**.



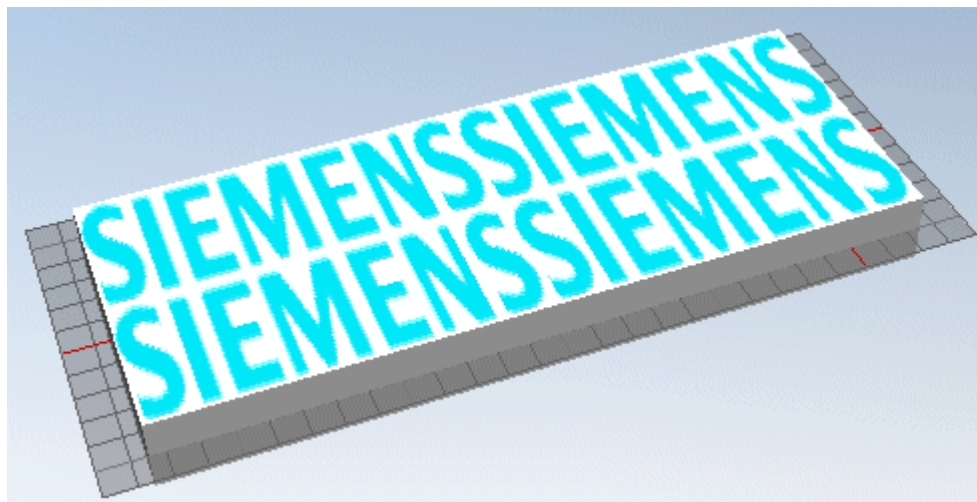
- You can also show the plate with a different **Orientation**:

Orientation	Looks like this
Floor	
Front wall	
Side wall	

- When you select **Show on Both Sides**, 3D shows the image on the top/front and on the bottom/back of the inserted plate.



- When you select **Fit Image Size**, and enter 2 tiles for the x-dimension and the y-dimension, 3D inserts two tiles of the image in each dimension.



- To move the textured plate to another position of your choice, click it with the left mouse button and drag the mouse.
- To move the textured plate up on the base plate, hold down **Ctrl** and press the **Up** arrow.
- To change the **size** of the textured plate after you inserted it, select it and click **Edit 3D Properties** > **Tab Transformation** on the **Home** ribbon tab. Then change the **scale** according to you needs.
- To set if the textured plate is an **obstacle for the worker**, select it. Then, click **Edit 3D Properties** on the **Home** ribbon tab and click the **Tab Graphic Settings**.
- To assign a material to the textured plate, select it and then click **Edit 3D Properties** > **Tab Material** on the **Home** ribbon tab.

Note

When you start to insert a newly created shape into an external graphic group, *3D* automatically activates **Show External Graphic Groups**  until you click the destination position so that you can align the new graphic at the mouse pointer with all other graphics of the destination graphic group.

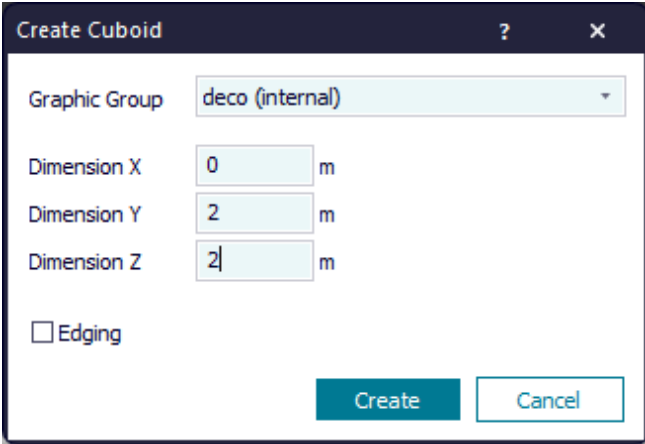
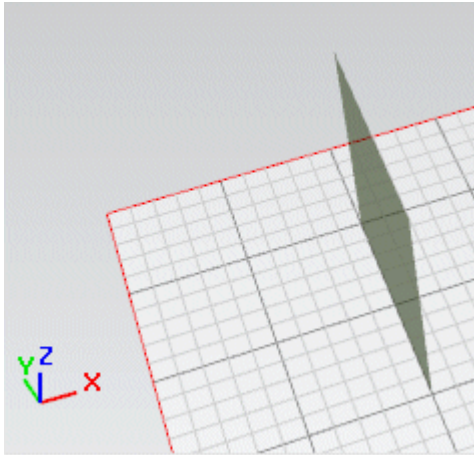
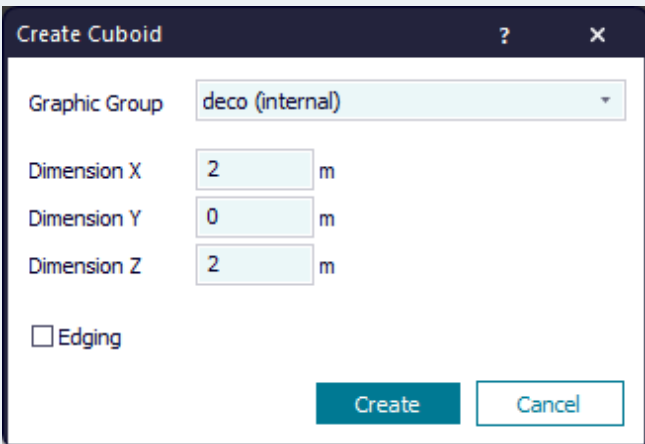
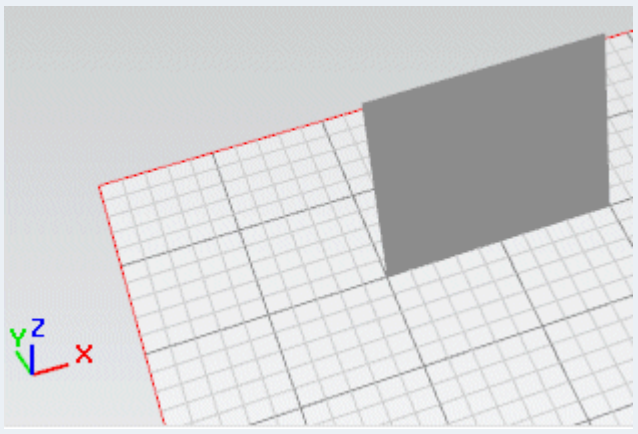
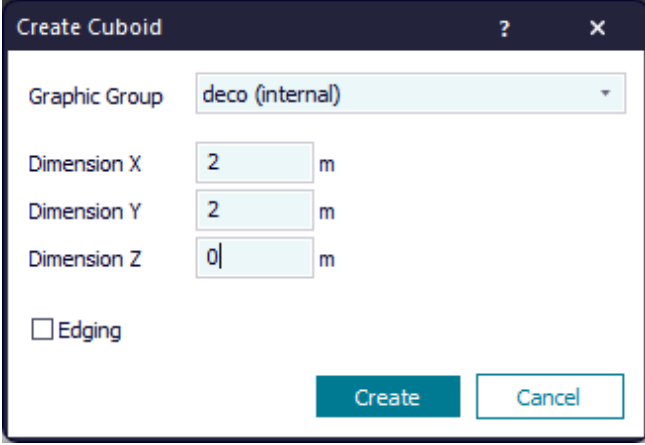
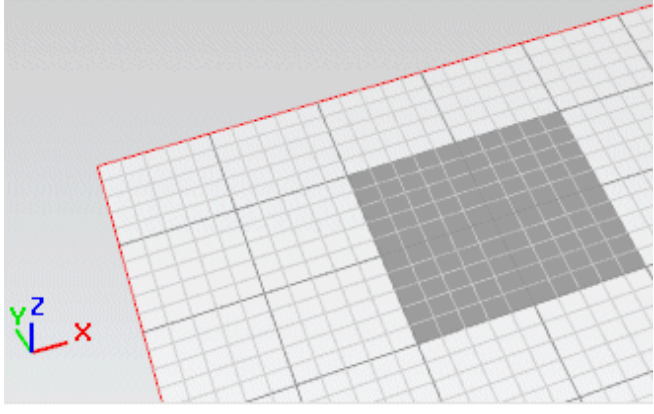
Back to [Create Your Own Objects](#)

Create a Non-textured Plate [example]

You can create a non-textured plate in your simulation model.

Proceed as follows:

- Click  **Cuboid** on the **Edit** ribbon tab.
- Enter 0 into the respective text box, to create the plate on this plane.

Setting	Looks like this
	
	
	

- Click **Create** and proceed as described under [Create a Textured Plate](#) to place or move the plate.

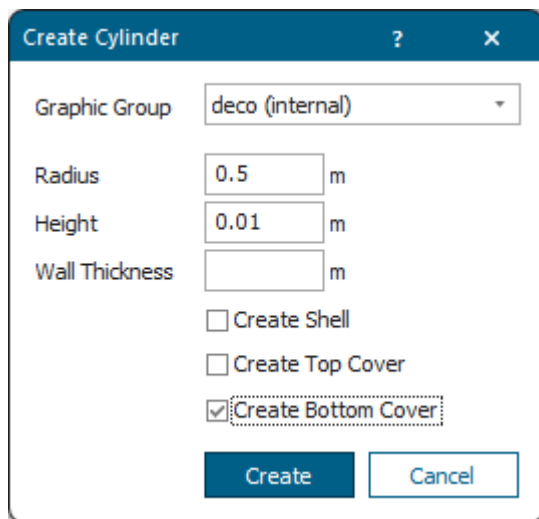
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Model a Bottle

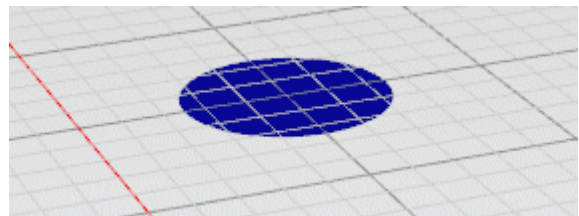
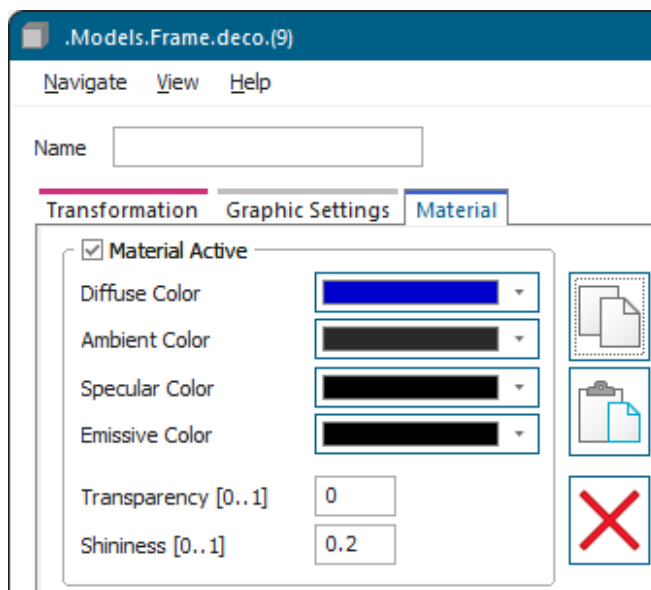
With the 3D shapes **Cylinder** and **Cone Frustum** we can put together a bottle or a similar-looking item. To better distinguish the different components of the bottle when putting them together, use easily distinguishable colors.

First, we model the bottom of the bottle with a cylinder:

- Click **Cylinder**  **Cylinder**.
- We want to create a flat bottom of the bottle. To do so, we typed in and selected these settings:

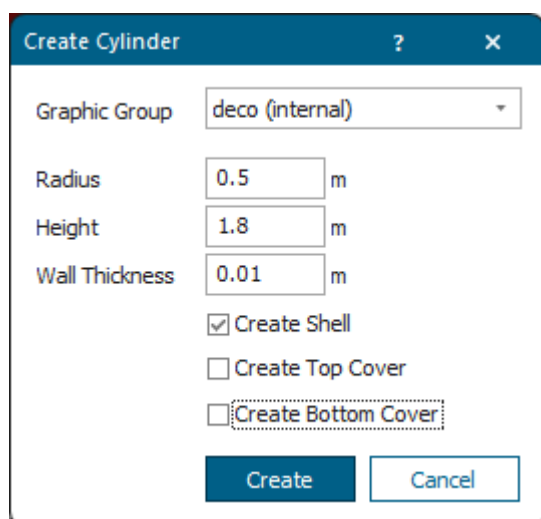


- After inserting it, our flat cylinder is gray by default. As this is hard to recognize on the gray grid plate and the gray background of the *Frame*, we assigned it another color.

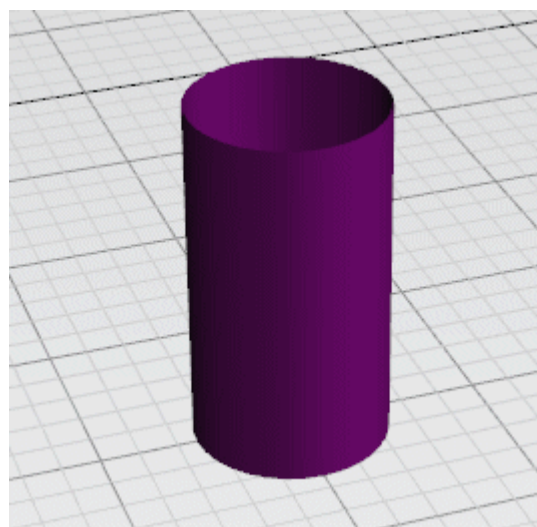
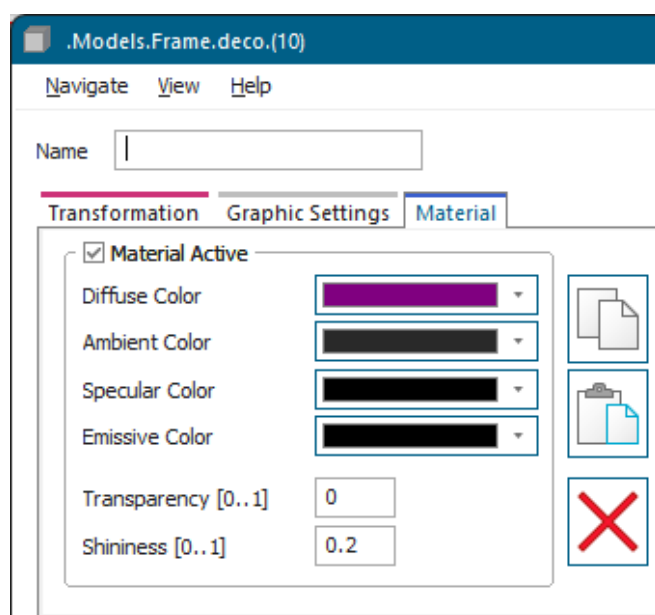


In the second step we model the body of the bottle. For this use a **Cylinder** as well.

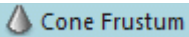
- Click **Cylinder**. Our body of the bottle is a cylinder that is open at the bottom and at the top with these settings:

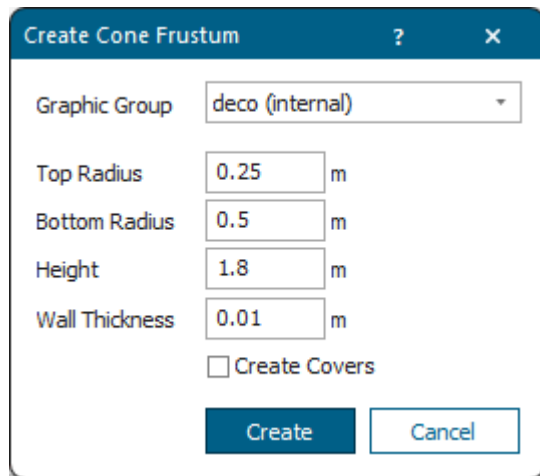


- After we assigned the inserted cylinder another color, it looks like this:

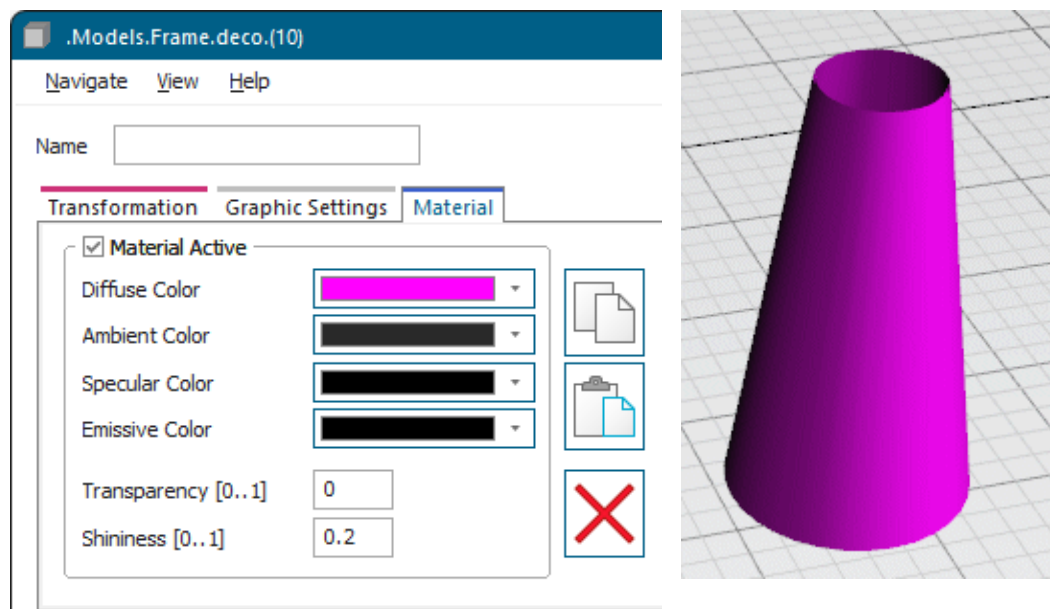


In the third step we model the part of the bottle that gets narrower toward the top, the bottleneck:

- Click **Cone Frustum** . We used these settings:

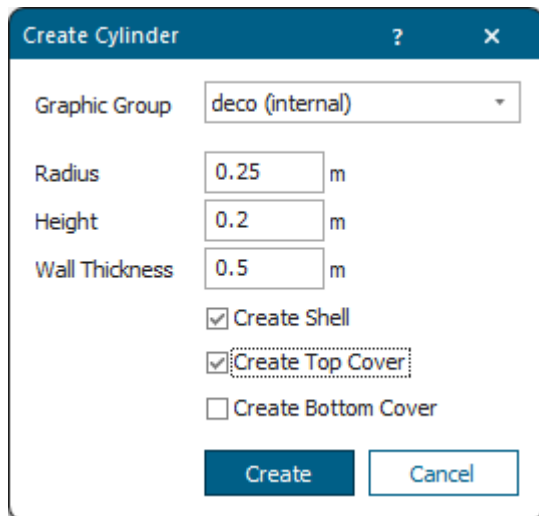


- After we assigned the inserted cone frustum another color, it looks like this:

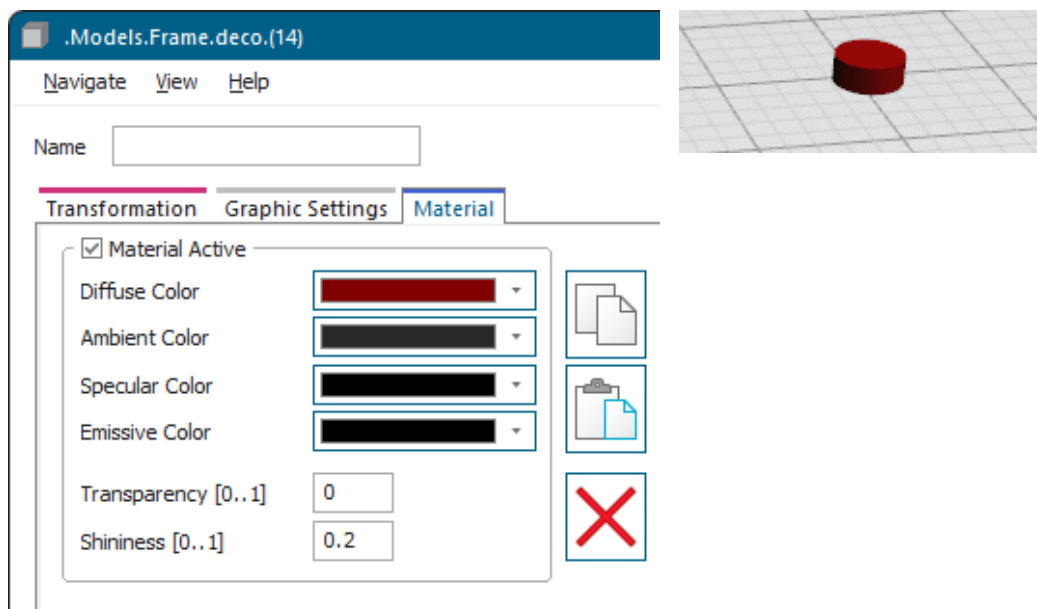


In the fourth step we model the cap of the bottle. For this we use a **Cylinder**:

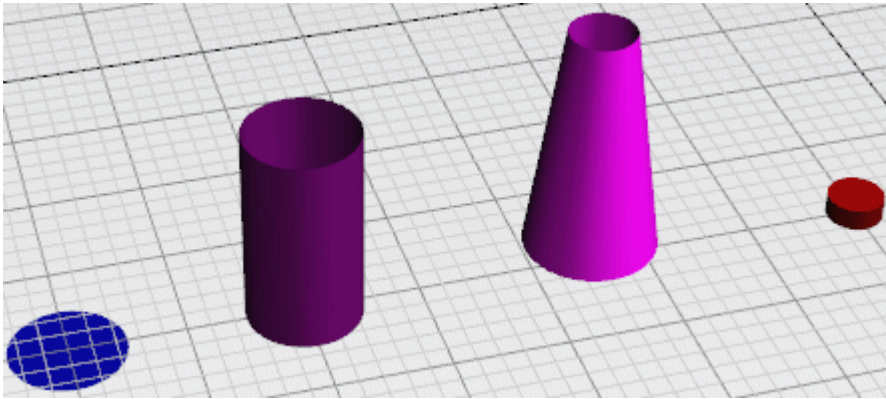
- Click **Cylinder**. To make the cap fit the neck of the bottle, it has to have the same radius of the neck, which we designed above. We used these settings:



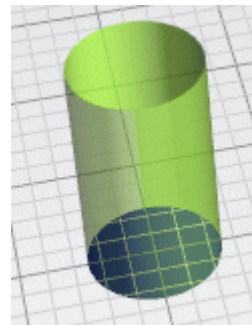
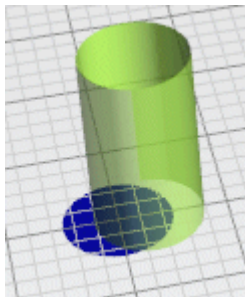
- After we assigned the inserted cylinder another color, it looks like this:



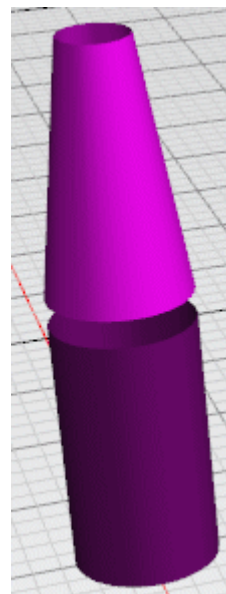
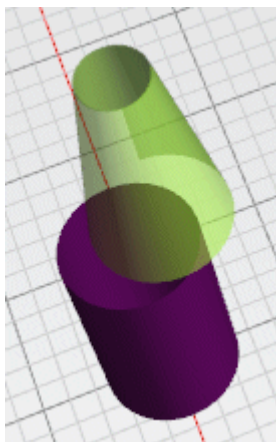
The individual parts of our bottle look like depicted below. Now we just have to put the bottle together.



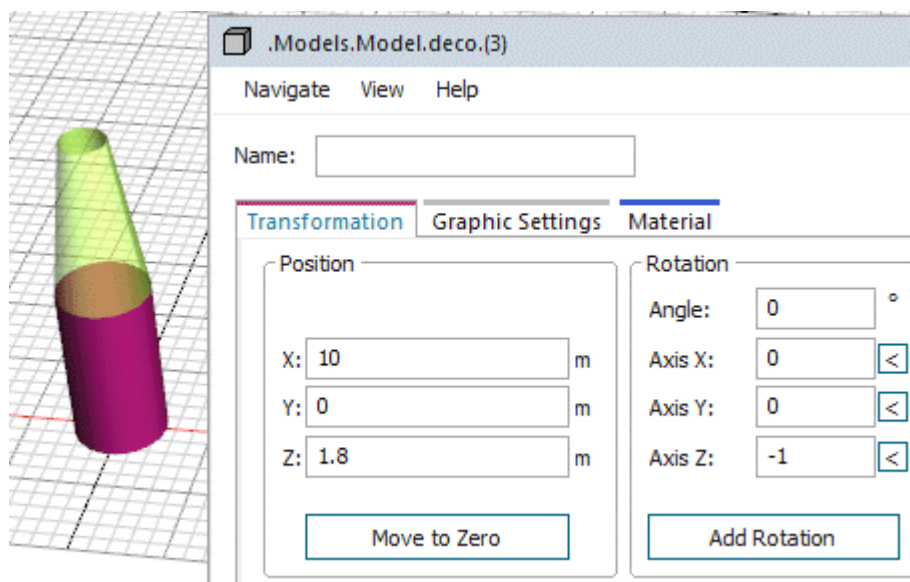
- First, we put the **body of the bottle** onto the **bottom of the bottle**. To do so, select the **body of the bottle** and drag it fairly precisely over the **bottom of the bottle**. Then click **Align to Grid**  on the **Edit** ribbon tab. Group these two parts by clicking **Group** on the **Edit** ribbon tab.



- Next, we place the **bottleneck**. To do so, select the bottleneck. Drag it fairly precisely over the bottom part of the bottle. Then click **Align to Grid** .

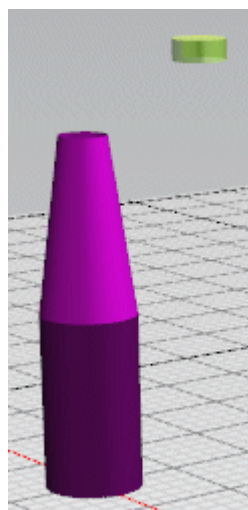


Now these two parts are located directly above each other, the upper part does not sit flush on the lower part though. To fix this, we have to move the bottleneck downward in the **Z-Dimension**. Click the bottleneck with the right mouse button and select **Edit 3D Properties**. Adjust the **Z-Dimension** until everything fits. We used the settings below.



Group the parts of the bottle.

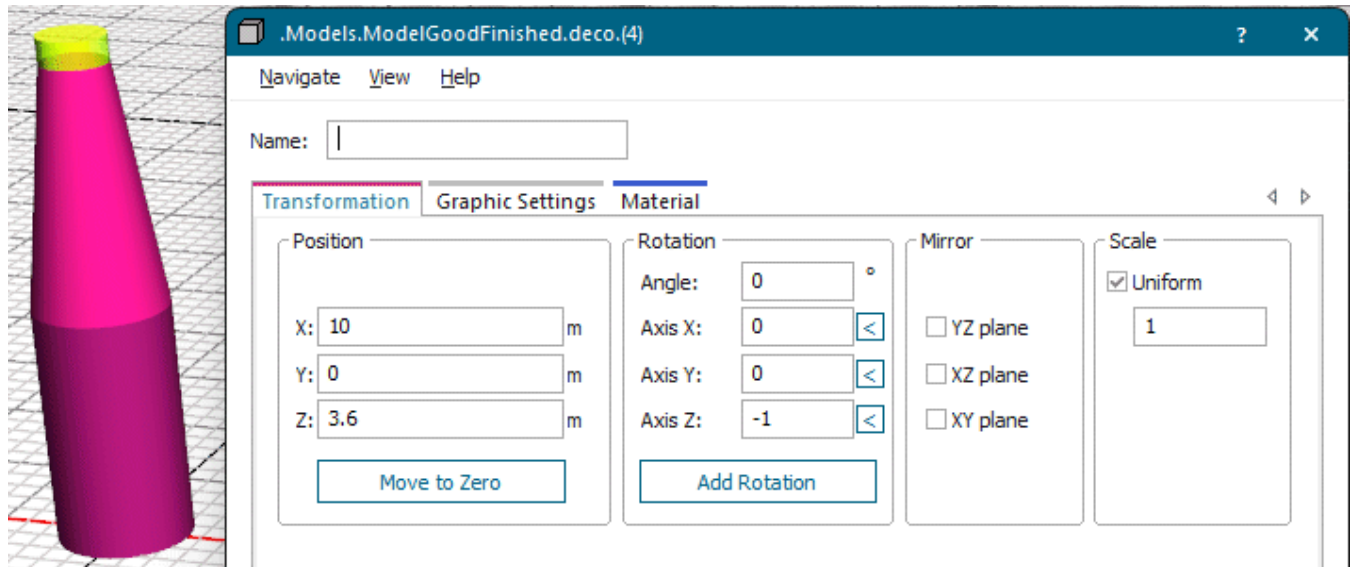
- In our last step we place the **cap** onto the **bottleneck**. To do so, select the cap. Hold down **Ctrl** and press the **Up arrow** on the keyboard to move it upward over the neck of the bottle. Press the **Left arrow** to place the cap fairly precisely over the neck of the bottle.



Then, click **Align to Grid** . Now the cap is located directly above the bottleneck, and we just have to move it down.



- Adjust the **Z-dimension** until everything fits. We used these settings:



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Set How Plant Simulation Shows an Object

Depending on which level of the object hierarchy you open a window, an object can look different.

For this reason we differentiate between the **outside representation** of the object and the **inside representation** of the object:

- The **outside representation** of a *Frame* designates the entirety of all graphics, which will be selected when you click the *Frame* in a window that opened the location of this *Frame*.

This includes:

- Its graphic groups, which are marked as **Visible [graphic group]** and **External [graphic group]**.
- Contained objects, which are not explicitly excluded from the parent outside representation, i. e., for which you selected **Exclude From "Show content" of the Location**. This applies only if you selected **Show Content** for the *Frame*.
- The **outside representation** of other objects includes the following:
 - Its graphic groups, which are marked as **Visible [graphic group]**.
 - Its state graphics, which you set according to the **Orientation [state graphic]**.
 - Its **material flow direction** if the object is length-oriented.
- The **inside representation** of a *Frame* designates the entirety of all graphics, which are shown, when you open a *Frame* in a window.

This includes:

- Its graphic groups, which are marked as **Visible [graphic group]** and **Internal [graphic group]**.
- The **outside representation** of its contained objects and *Frames*.
- The **inside representation** of other objects includes:
 - Its graphic groups, which are marked as **Visible [graphic group]**.
 - Its state graphics, which you set according to the **Orientation [state graphic]**.
 - The **outside representation** of its contained objects.

When editing the object, it might be helpful to show the **outside representation** and the **inside representation** at the same time. To do so, activate **Show External Graphic Groups** in a window, which contains the object you are editing.

In the active window, you can, in addition, show or hide the following:

- All *Connectors*, *Interfaces*, and *Markers* by clicking the button **Show Connections**.
- The material flow direction arrows on all length-oriented objects by clicking the button **Show Material Flow Directions [button]**.

Compare **Work with Graphics**

Compare **What Plant Simulation Shows in a Window**

Back to **Work with Objects**

What Plant Simulation Shows in a Window

The *Plant Simulation* window shows graphic groups, graphics, sensors, and object states depending on different circumstances.

What *Plant Simulations* shows in a window depends on the content of the object or of the *Frame*, on the settings of this content, and on settings of the window itself:

Plant Simulation shows	if	and	or/and
An object (excluding <i>Frame</i> , <i>connection</i> (<i>Connector</i> , <i>Interface</i> , <i>Marker</i>), or <i>sensor</i>)	its location is open in the window	at least one of its graphic groups is Visible and External and contains graphics	—
A <i>Frame</i>	its location is open in the window	either at least one of its graphic groups is Visible and External and contains graphics	or the <i>Frame</i> shows Content and contains at least one object or <i>Frame</i> that is not Excluded From Show Content .
An External graphic group of a <i>Frame</i>	the graphic group is Visible	the <i>Frame</i> is shown in the window	or the <i>Frame</i> is open in the window, and the window shows External Graphics .
An Internal graphic group of a <i>Frame</i>	the graphic group is Visible	the <i>Frame</i> is open in the window	—
A graphic group of an object	the graphic group is Visible	either the object is open in the window	or the object is shown in the window.
The Name of a <i>Frame</i> or object	the object shows Captions	the <i>Frame</i> /object is shown in the window	and the window shows Names .
The Label of a <i>Frame</i> or object	the object shows Captions	the <i>Frame</i> /object is shown in the window	and the window shows Labels .
The state graphics of an object	the object shows States	either the object is open in the window	or the object is shown in the window.
The <i>Connector</i> of a <i>Frame</i>	the window shows Connections	either the <i>Frame</i> is open in the window	or the <i>Frame</i> is shown in the window, and the <i>Frame</i> shows Content , and both connected objects are not Excluded From Show Content .

Plant Simulation shows	if	and	or/and
The <i>Interface</i> or <i>Marker</i> of a <i>Frame</i>	the window shows Connections	either the <i>Frame</i> is open in the window	or the <i>Frame</i> is shown in the window, and the <i>Frame</i> shows Content , and the <i>Interface/Marker</i> is not Excluded From Show Content .
A sensor of a length-oriented object	the length-oriented object shows Sensors	either the length-oriented object is open in the window	or the length-oriented object is shown in the window.
Material Flow Directions of a length-oriented object	the window shows Flow Directions	the length-oriented object is shown in the window	—

See also

[Set How Plant Simulation Shows an Object](#)

[Graphic Groups \[described\]](#)

Using Alternative Graphic Groups

Use Alternative Graphic Groups

In this sample model we show how to model a simplified airplane and how to show the progress of the airplane assembly. To do so, we show the selected graphic groups which graphically display the finished components.

We will solve the following modeling tasks:

- [Create a New MU Class for the Object](#)
- [Create Alternative Graphic Groups for the Components](#)
- [Show the Progress of the Airplane Assembly](#)

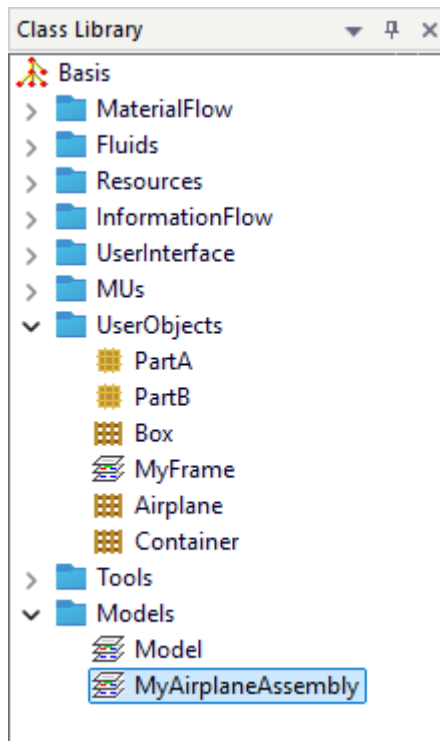
Back to [Work with Graphics](#)

Create a New MU Class for the Object

We first create a new MU class for the object for our alternative graphic group.

First, we will create a new MU class that represents the airplane we are going to build:

- Open the folder *MUs* in the *Class Library*.
- Click the object *Container* with the right mouse button and select **Duplicate**.
- Press **F2** and rename the newly created *Container* to *Airplane*.



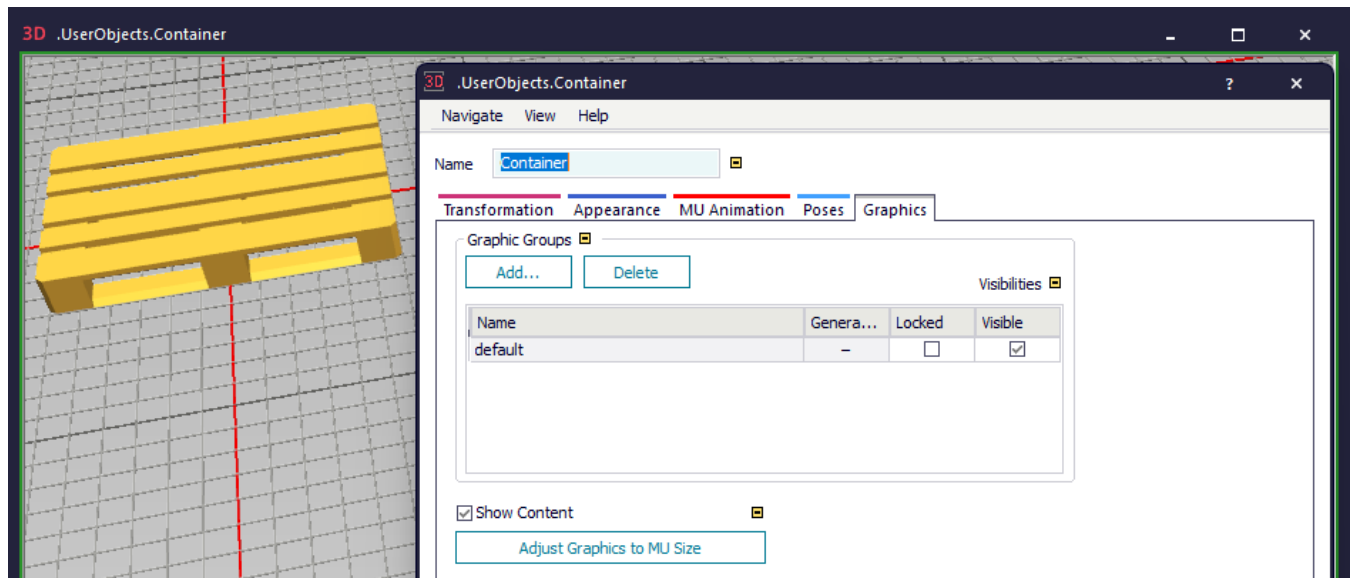
While you're at it, you can also rename the *Frame* named *Models* in the folder **Models** to *MyAirplaneAssembly*.

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Create Alternative Graphic Groups for the Components

After creating the MU class *Airplane* for our model, we will create the graphics of the different components of the airplane. For demonstration purposes we will content ourselves with the **Shapes** that *Plant Simulation* provides. In a real situation, we would naturally use the CAD drawings we received from the engineering department.

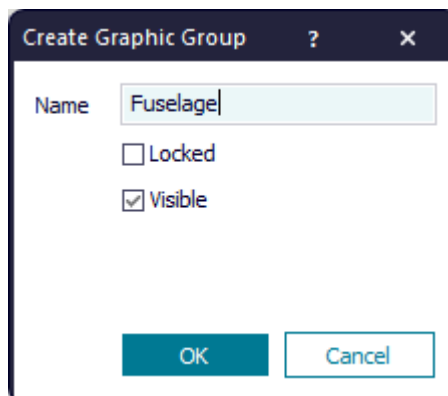
- Open the *Frame* containing our simulation model, *My3DAirplaneAssembly* in our case.
- Edit the MU class *Airplane*. Click it in the *Class Library* with the right mouse button and select **Open in 3D**. Click into the background of the *Frame* **.UserObjects.Airplane** with the right mouse button and select **Edit 3D Properties**.



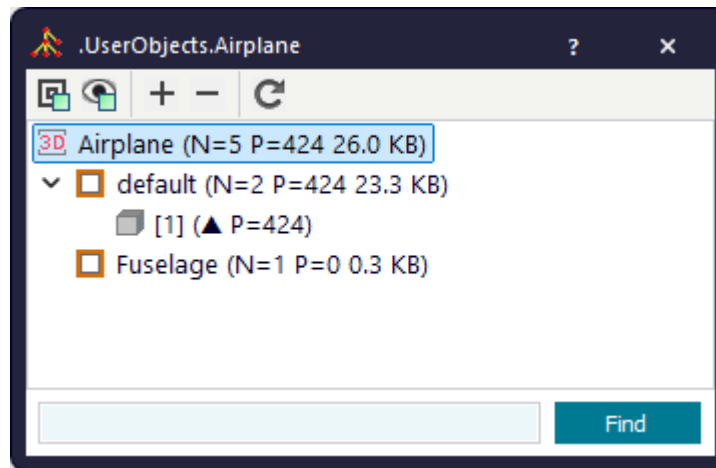
Note

As we do not need the default graphic of our airplane, the pallet with the name **default**, we delete this graphic by selecting it and pressing **Del**.

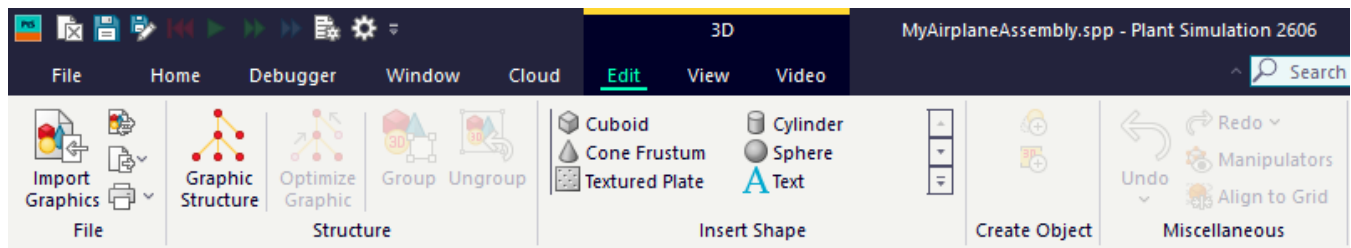
- Click **Add** to add a new **graphic group**. We first want to add the graphic group for the **fuselage** of the airplane. Type Fuselage into the text box **Name**.



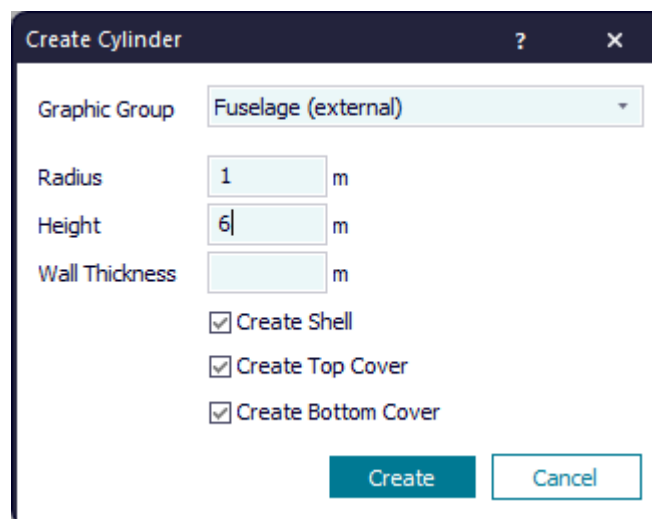
Once you clicked **OK** in the dialog above and **Apply** in the dialog **Edit 3D Properties**, the dialog **Show Graphic Structure** shows our new graphic group as well.



- Now we **create** the **actual graphic** that represents the **fuselage**. Change to the **Edit** ribbon tab and select **Cylinder**.



- Select the graphic group to which you want to add the newly created graphic, **Fuselage** in our case. Enter the dimension of the cylinder and click **Create**.

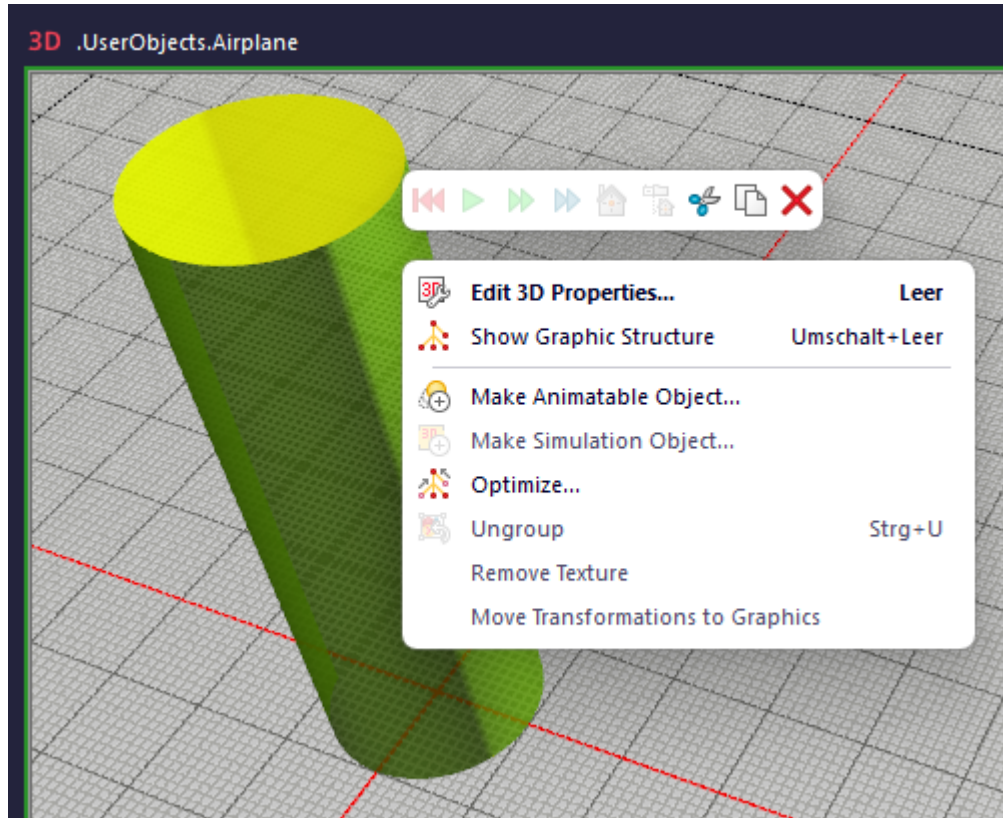


Plant Simulation attaches the cylinder to the cursor in the 3D window **.UserObjects.Airplane**. Drag it to the position of your choice and click the left mouse button to insert it.

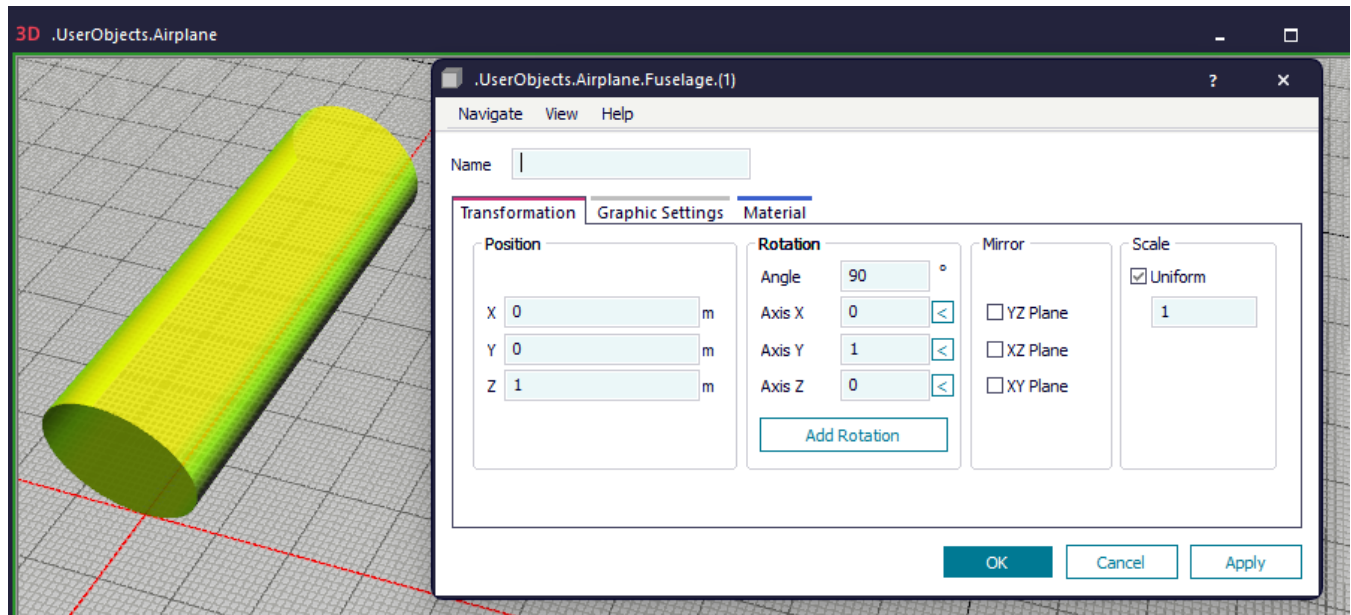
Note

Make sure that **Scale automatically** on the tab **Transformation** is cleared, otherwise *Plant Simulation* will not show the cylinder with the settings you entered.

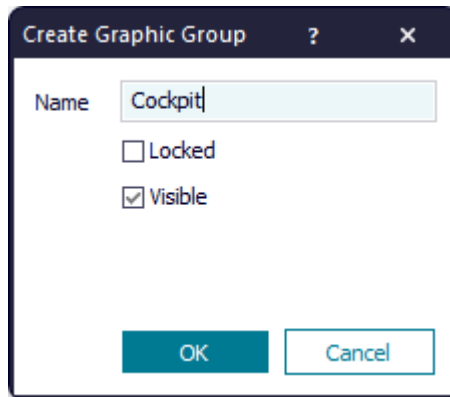
- 3D inserts the cylinder in the upright position. This is not what we want though. We need our fuselage to be in the horizontal position. To rotate it by 90 degrees, click it with the right mouse button and select **Edit 3D Properties**.



- Enter the settings from the screenshot below and click **Apply** to rotate the fuselage by an angle of 90 degrees on the y-axis.

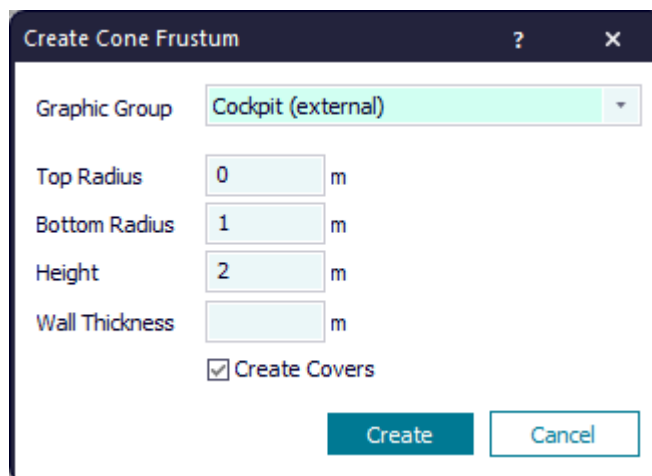


- In the next step, we are going to **add** the **graphic group** for the **cockpit** of our airplane. Click with the right mouse button in the *Frame* **.UserObjects.Airplane** and select **Edit 3D Properties**. Change to the tab **Graphics** and click **Add**. Type **Cockpit** into the text box **Name** and click **OK** in the dialog. Then click **Apply** in the dialog **Edit 3D Properties**.

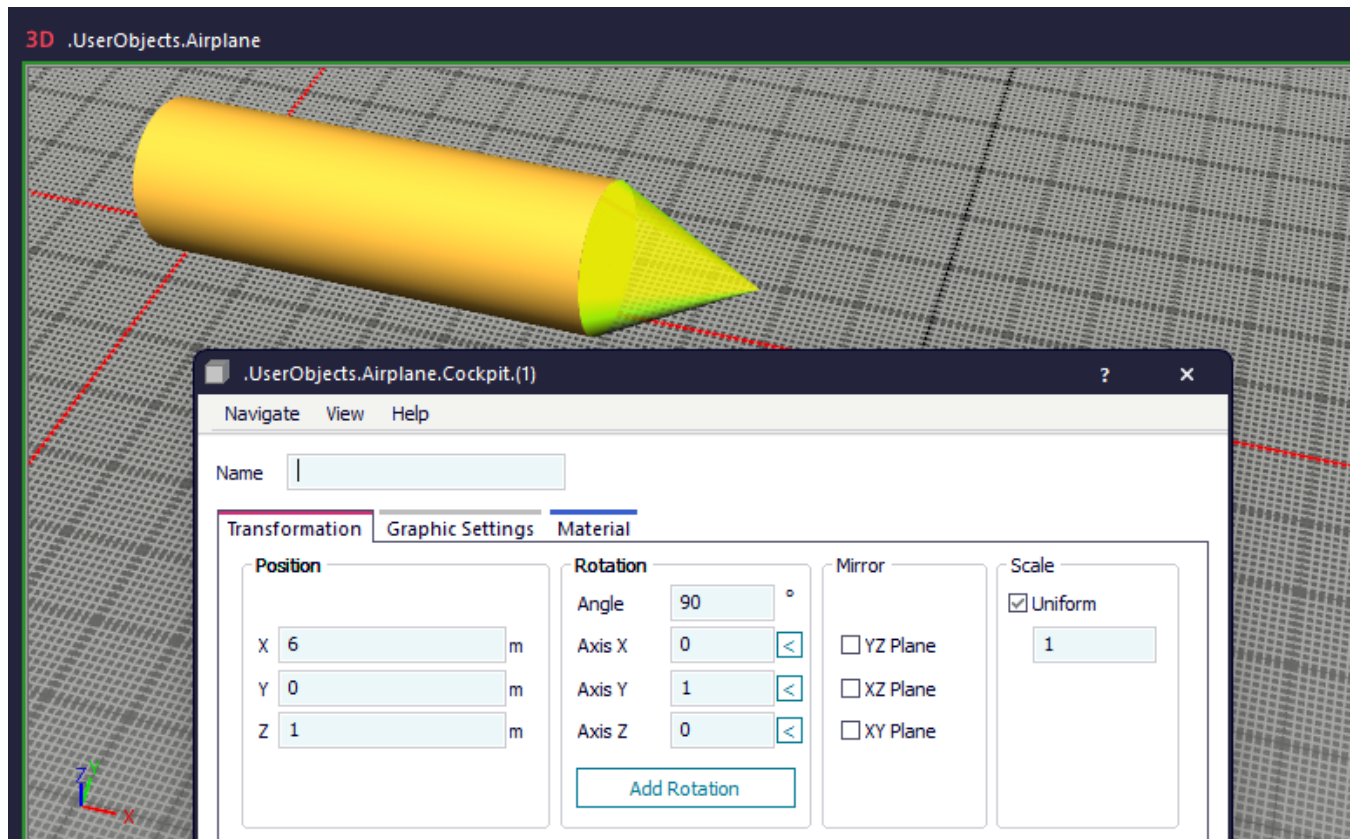


- Then, **create** the **actual graphic** that represents the **cockpit**. Change to the **Edit** ribbon tab and select **Cone Frustum**  **Cone Frustum**.

Select the graphic group to which you want to add the newly created graphic, **Cockpit** in our case. Enter the dimension of the cone and click **Create**. *Plant Simulation* attaches the cone to the cursor in the 3D window **.UserObjects.Airplane**. Drag it to the position of your choice and click the left mouse button to insert it.

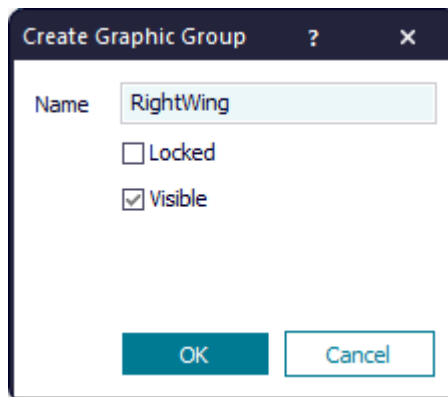


- Rotate the **Cockpit** by 90 degrees and place it at the right-hand side of the fuselage. We used the settings shown the screenshot below.

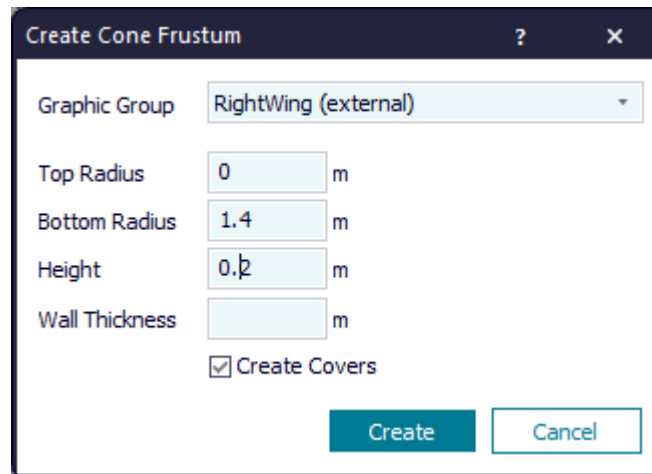


While you're at it, you might also assign the cockpit a color on the tab **Material**.

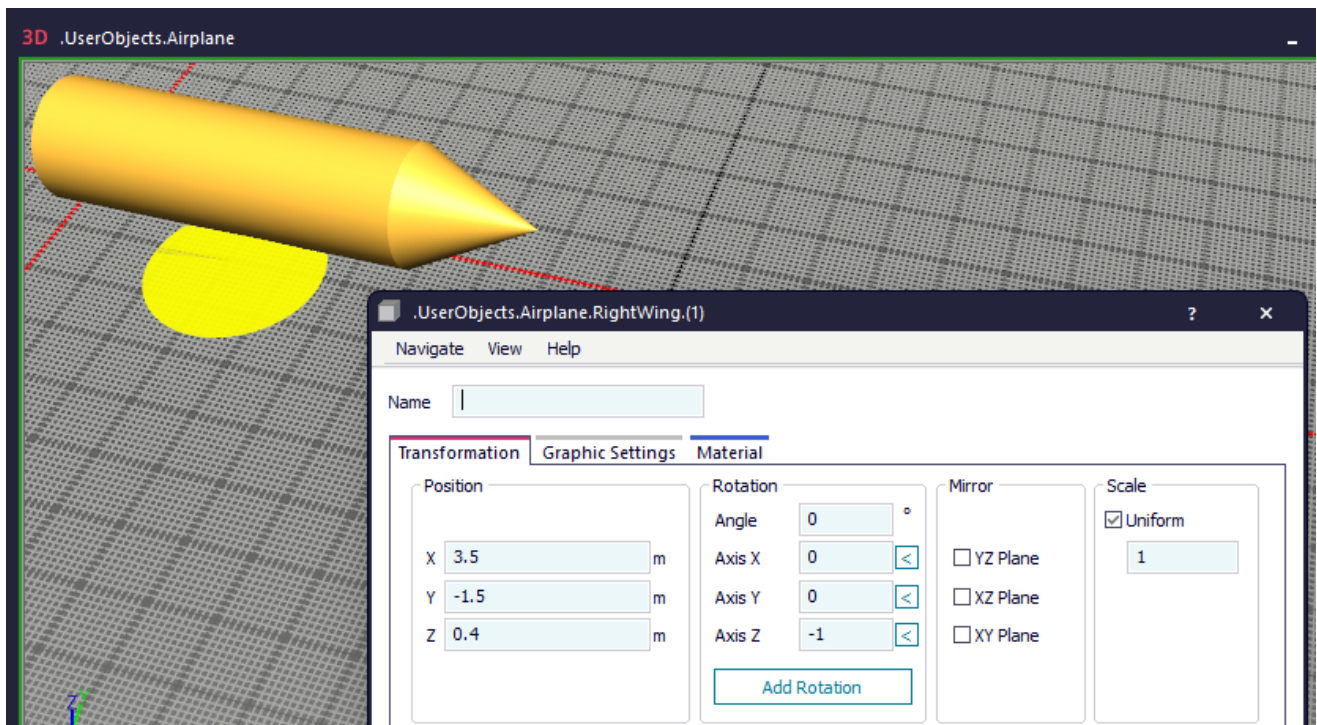
- As our airplane also needs a **right wing** and a **left wing**, we create these next. Repeat the steps described above.
 - **Add** the **graphic group** for the **right wing**.



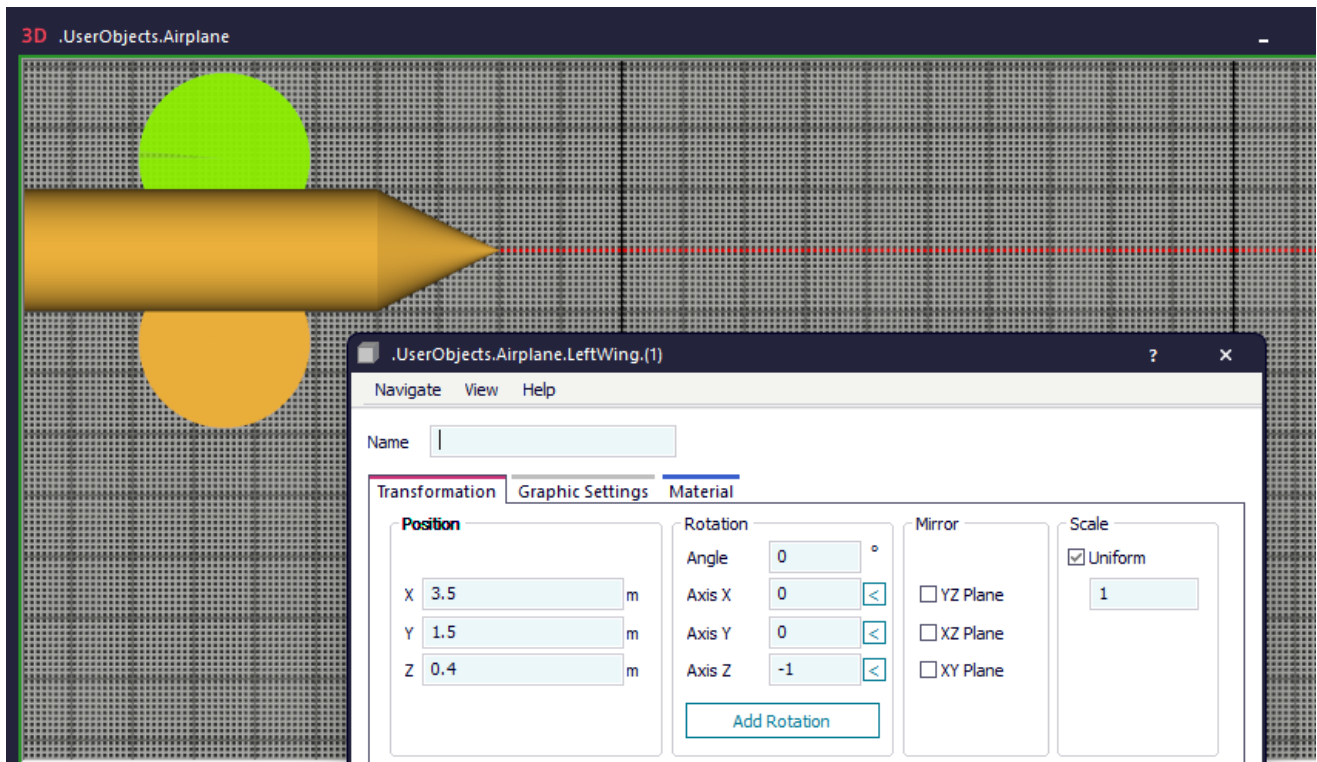
- Select the graphic group **RightWing** to which the graphic will be added. **Create** the **actual graphic** for the **right wing**.



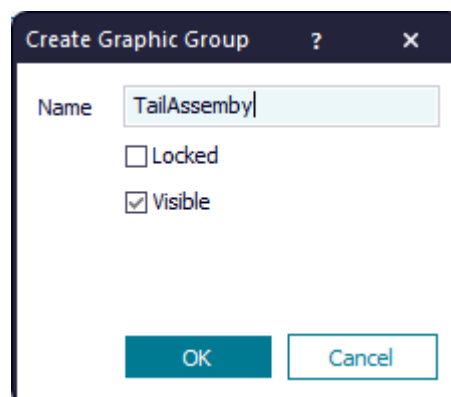
- Place the right wing at the appropriate position on the fuselage. We used the settings below.



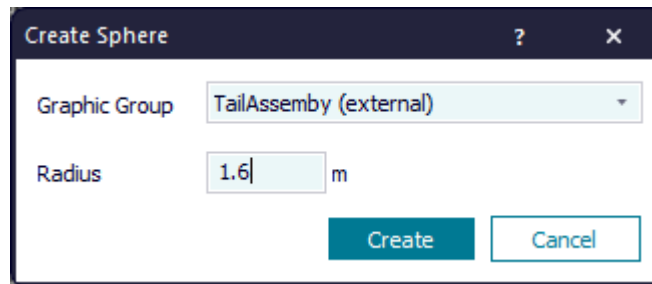
- Repeat these steps for the **left wing**. We used the settings below to place the left wing at the appropriate position on the fuselage.



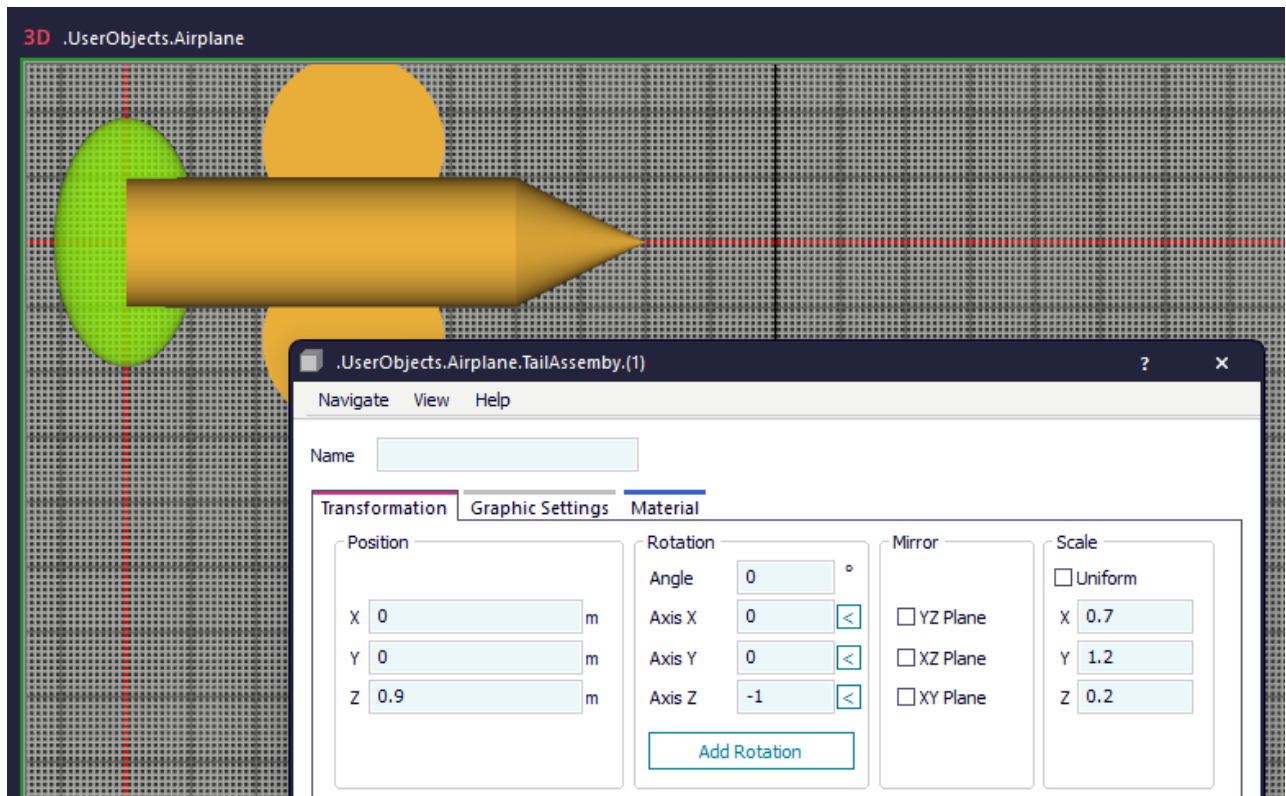
- In our last step we are going to create the **tail assembly** (empennage) consisting of the **vertical stabilizer** and the **horizontal stabilizer** (tailplane). Repeat the steps described above.
- ◊ Add the **graphic group** for the **tail assembly**.



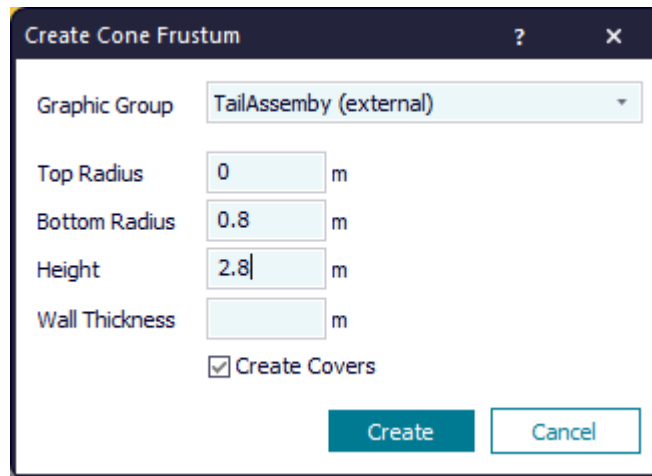
- ◊ Select the graphic group **TailAssembly** to which the graphic will be added. **Create the actual graphics** representing the **tail assembly**. We used a **Cone** for the **vertical stabilizer** and a **Sphere** for the **horizontal stabilizer**.



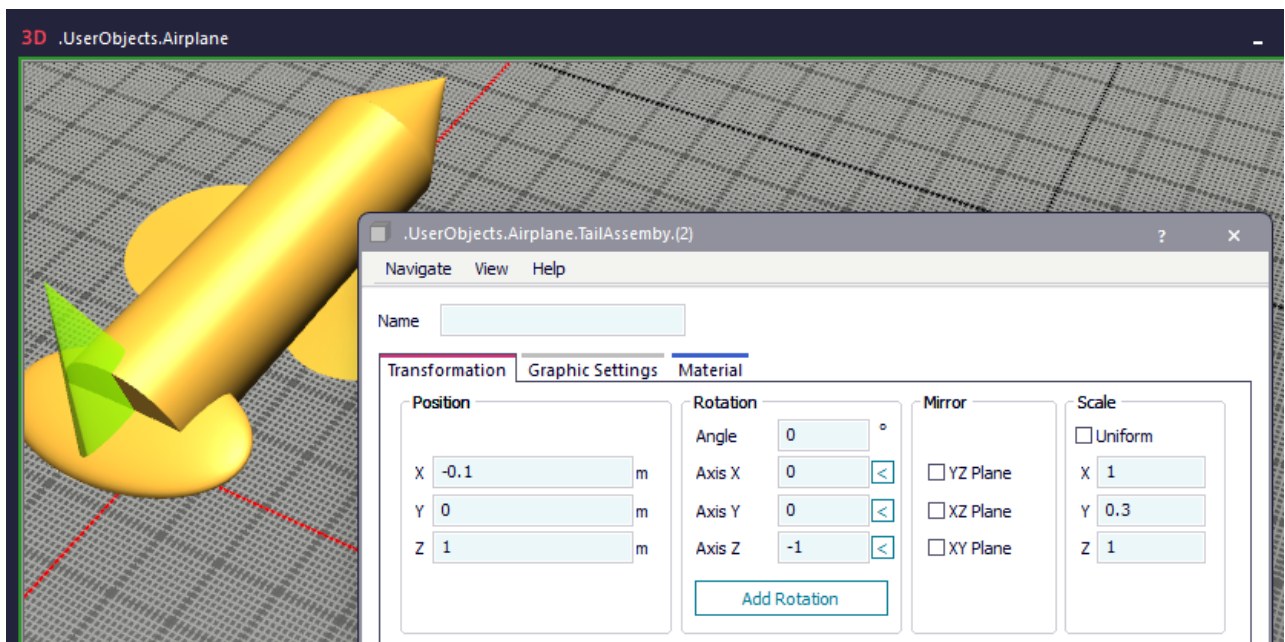
- ◆ Place the **horizontal stabilizer** at the appropriate position on the fuselage. We used the settings below. We also **scaled** the sphere in all three dimensions to stretch and to flatten it.



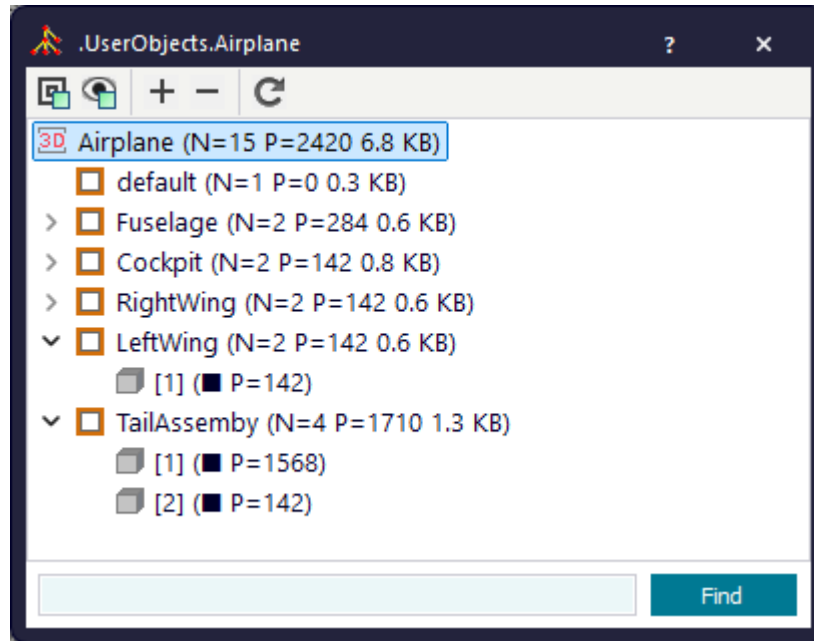
- ◆ Select the graphic group **TailAssembly** to which the graphic will be added. **Create** the **graphic** representing the **vertical stabilizer**.



- ◆ Place the **vertical stabilizer** at the appropriate position on the fuselage. We used the settings below. We also **scaled** the cone in the y-dimension.



The dialog **Graphic Structure** also shows that our tail assembly consists of two graphics.



Instead of using shapes which we create in 3D itself, we can also [Import JT Graphics Representing an Object](#).

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Displaying the Progress of the Airplane Assembly

Show the Progress of the Airplane Assembly

In our last step we are going to show the progress of the airplane assembly in our plant by successively hiding and showing graphic groups as the airplane moves from processing station to processing station.

We will demonstrate how to:

- [Insert and Configure the Source and the Processing Stations](#)
- [Program the Visibility Controls](#)
- [Change Orientation and Position of the Object on the Stations](#)

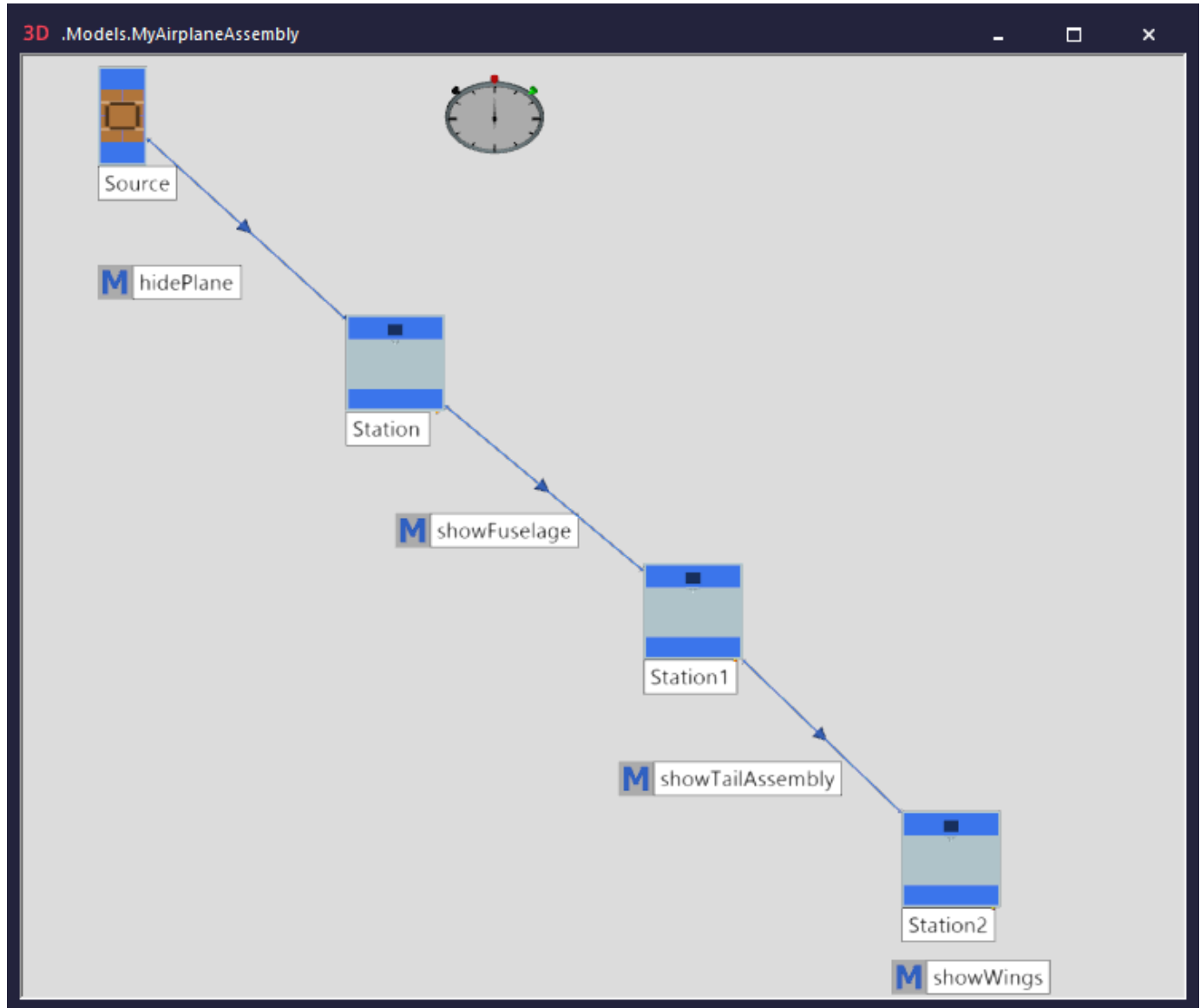
[Back to Work with Graphics](#)

Insert and Configure the Source and the Processing Stations

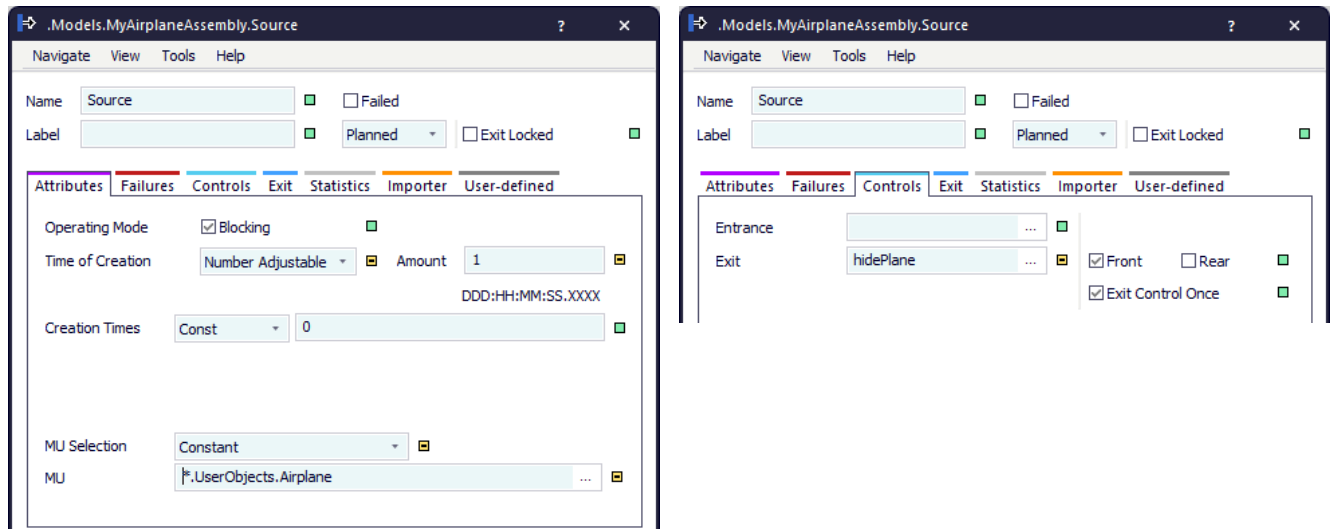
First, we are going to insert the *Source* that produces the airplane and the *stations* that process and assemble it. Connect the stations with *Connectors*.

While you are at it, you might also insert a *Method* below each of the stations and name them as shown in the screenshot. We will then program the individual methods in the next step.

Our finished model looks like this:



- Insert and configure the *Source* so that it matches the screenshots below. Make sure that it produces a **MU** of type **Airplane**. Select the method *hidePlane* as the **Exit Control** on the tab **Controls**.



- Insert and configure the *Station*. Enter a **Processing Time** of 10 seconds (0:10). Select the method *showFuselage* as the **Entrance Control** on the tab **Controls**.
- Insert and configure next station, called *Station1*. Enter a **Processing Time** of 10 seconds. Select the method *showTailAssembly* as the **Entrance Control** on the tab **Controls**.
- Insert and configure final station, called *Station2*. Enter a **Processing Time** of 10 seconds. Select the method *showWings* as the **Entrance Control** on the tab **Controls**.

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Program the Visibility Controls

Next, we are going to program the methods that we use as **Exit Control** and as **Entrance Control** of the stations for hiding and showing graphic groups as the airplane moves from station to station.

To do so, we will use the attribute **Visible**.

- Program the method *hidePlane*, which we use as the **Exit Control** of the *Source*. It **hides all components** of the airplane. Once all components are hidden, it moves the airplane on to the next station.

We entered the following source code:

```
@._3D.getGraphic("Fuselage").Visible := false
@._3D.getGraphic("Cockpit").Visible := false
@._3D.getGraphic("LeftWing").Visible := false
@._3D.getGraphic("RightWing").Visible := false
@._3D.getGraphic("TailAssembly").Visible := false
@.move -- move the airplane on to the next station
```

- Program the method *showFuselage*, which we use as the **Entrance Control** of the station named *Station*. It **shows the fuselage** and the **cockpit** of the airplane.

We entered the following source code:

```
@._3D.getGraphic("Fuselage").Visible := true
@._3D.getGraphic("Cockpit").Visible := true
```

- Program the method *showTailAssembly*, which we use as the **Entrance Control** of the station named *Station1*. It **shows** the **tail assembly** of the airplane.

We entered the following source code:

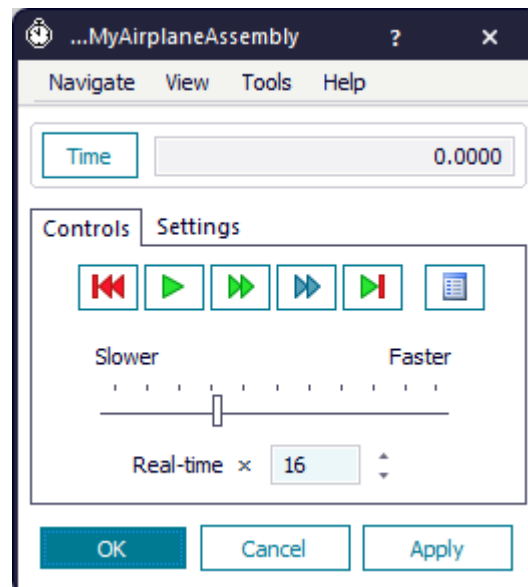
```
@._3D.getGraphic("TailAssembly").Visible := true
```

- Program the method *showWings*, which we use as the **Entrance Control** of the station named *Station2*. It **shows** the **wings** of the airplane.

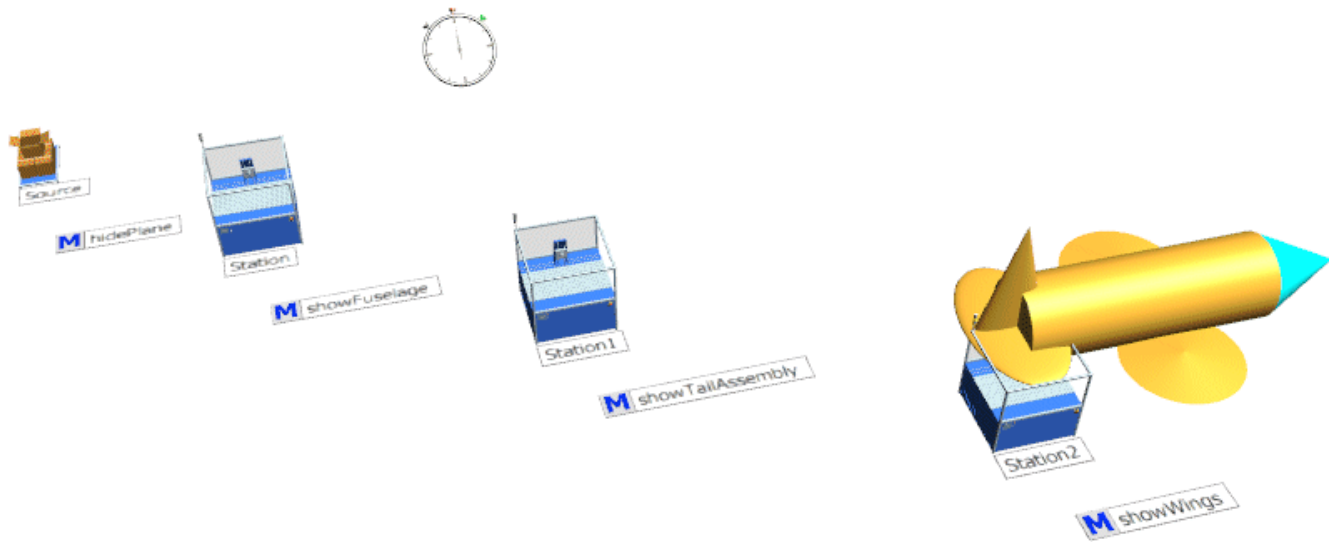
We entered the following source code:

```
@._3D.getGraphic("LeftWing").Visible := true
@._3D.getGraphic("RightWing").Visible := true
```

- As we are running the simulation, we want to see a smooth animation. A **Real-time** factor of 16 in the dialog of the *EventController* in our simulation model ensures this.



- Let's run the simulation to check if everything works as intended, which indeed it does.



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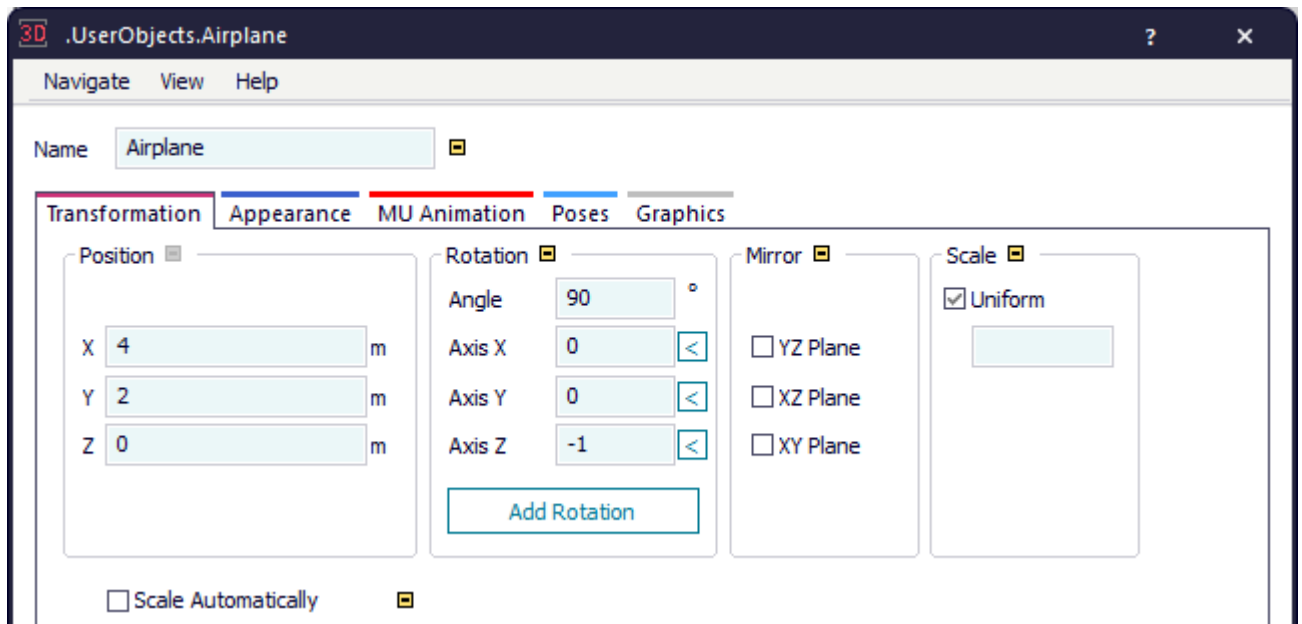
Change Orientation and Position of the Object on the Stations

You can adjust the orientation and the position the airplane on the stations. The default orientation of the airplane looks realistic on a conveyor, modeled for example with a *Conveyor*.

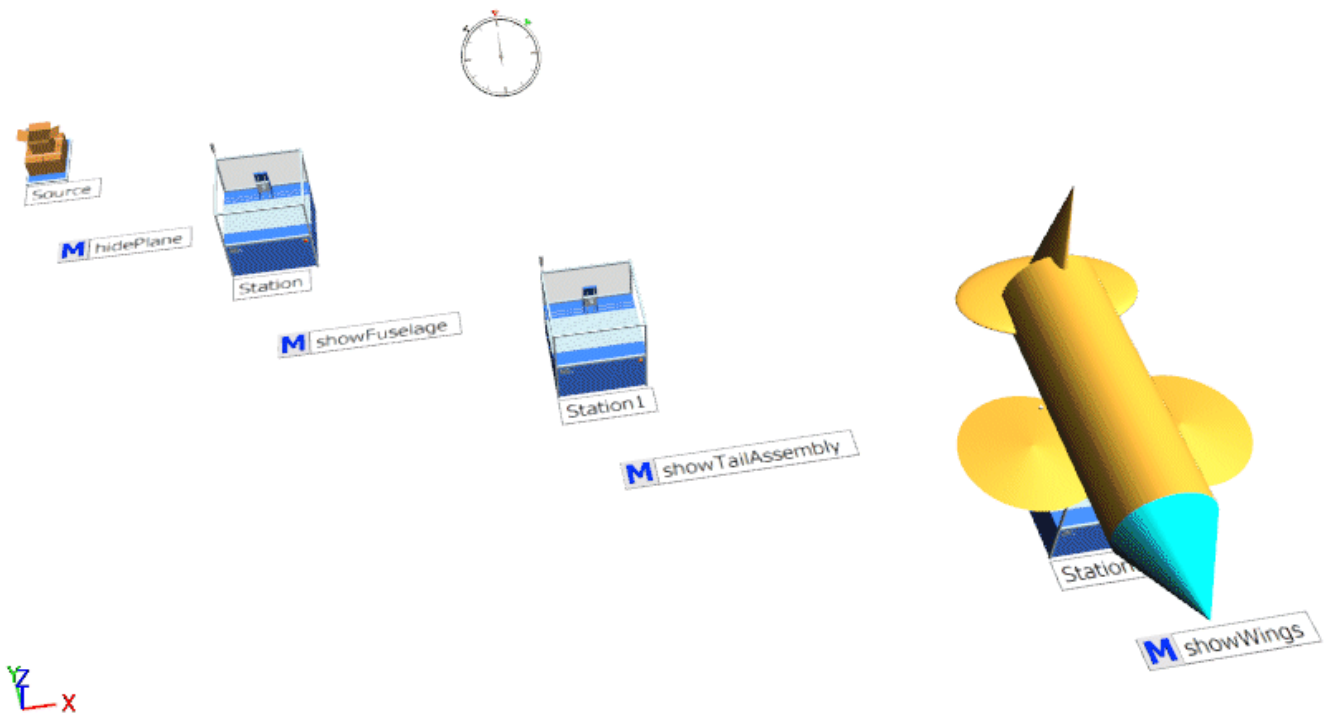
It does not look good on the *Stations* that we used though.

We are going to changes this:

- First, let's **rotate** the airplane by 90 degrees to the right. To do so, click the *class* of the airplane with the right mouse button, and select **Open in 3D**. Click the right mouse button in the background of the window and select **Edit 3D Properties**. Enter 90 into the text box **Angle** on the tab **Transformation > Rotation**.

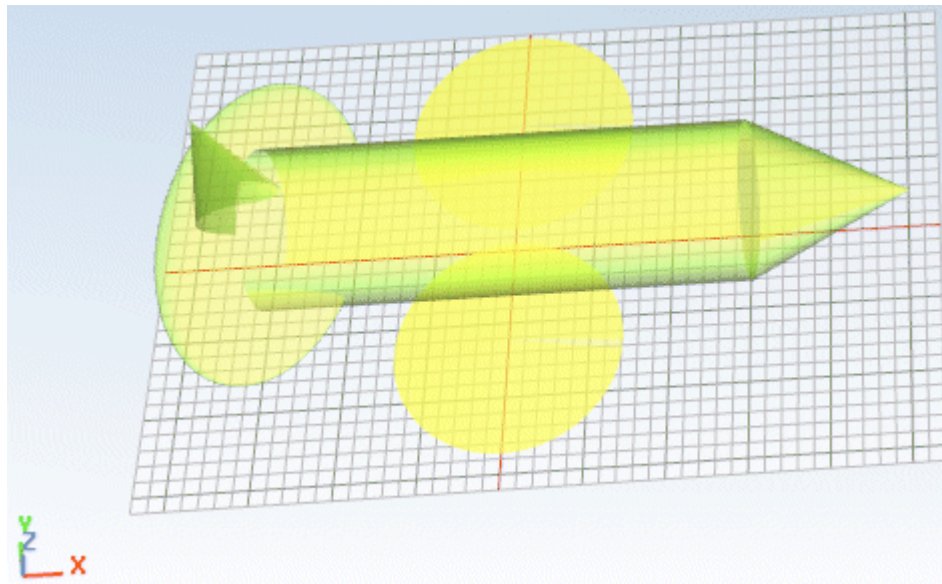


Run the simulation again to show the result of the rotation.

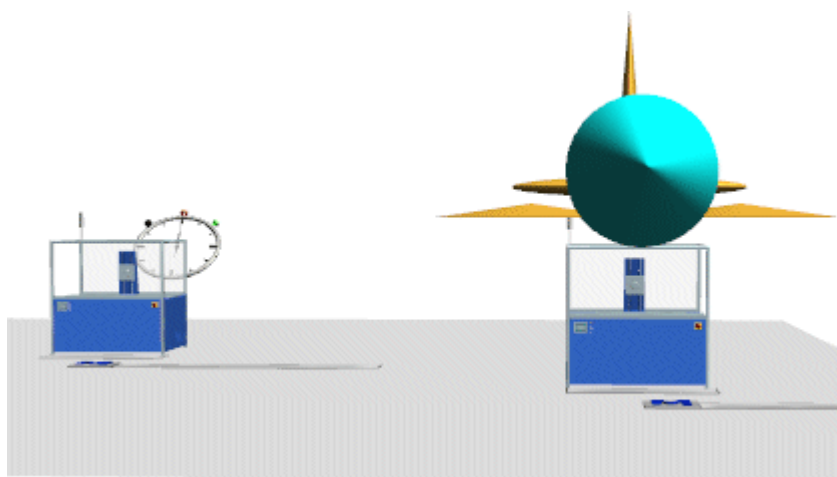


- Now let's change the **position** of the airplane on the stations. This is necessary, because by default *Plant Simulation* places the MUs at the origin of its scene, i.e., the point where the two red lines intersect. This is why the tail of our airplane is placed on the reference point of the stations.

To change the position of the airplane on the stations, open the *class* of the airplane in a new window. Press **Ctrl+A** to select all graphic groups of the airplane. Move them to the left with the **left arrow key** until the center of the wings is located on the y-axis. We also move the plane up on the z-axis by holding down **Ctrl** and **Shift** and pressing the **up arrow key**.



Run the simulation again to show the result of centering the airplane on the processing stations.



Back to [Show the Progress of the Airplane Assembly](#)

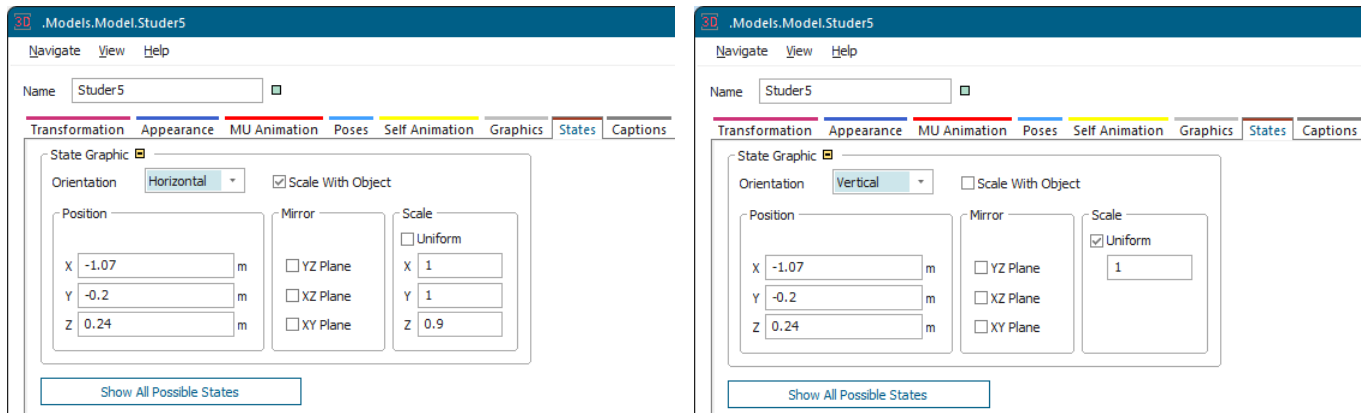
Displaying Object States

Show Object States

You can show the states of the objects in your simulation model with **State graphics**.

State graphics show the [States of the Material Flow Objects](#). A **state graphic** can show several states at the same time.

You can select the settings for the state graphic on the [Tab States](#) in the dialog **Edit 3D Properties**.



We will demonstrate how to:

- **Show States on a Signal Post Above the Machine**
- **Show States at the Front of the Machine**

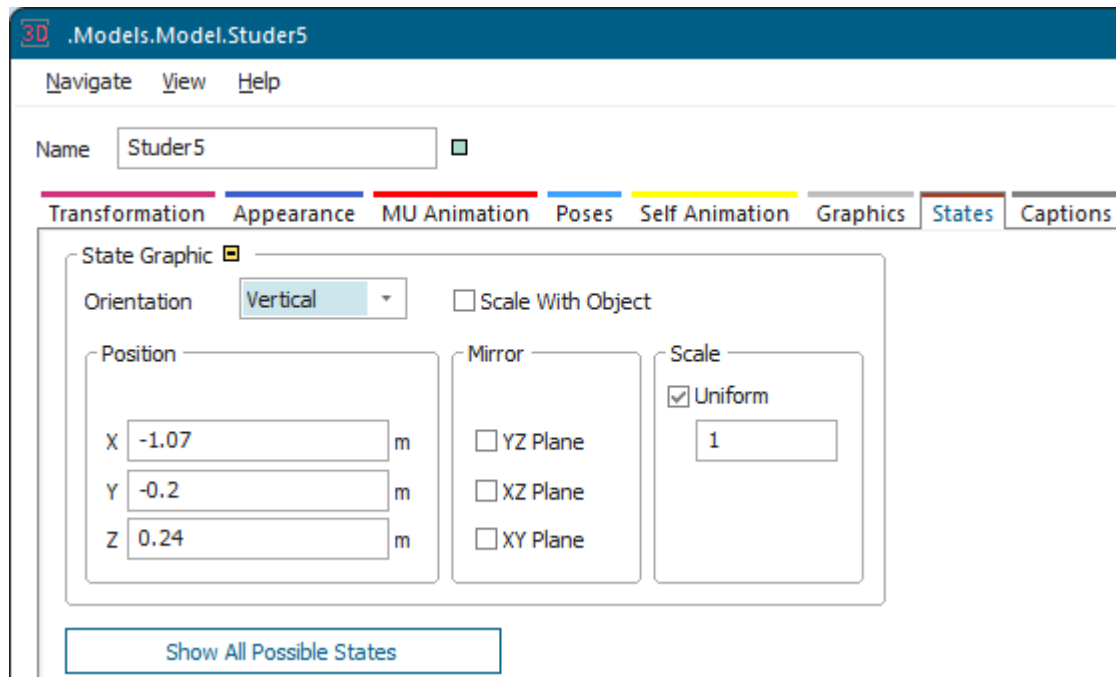
Back to [Work with Graphics](#)

Show States on a Signal Post Above the Machine

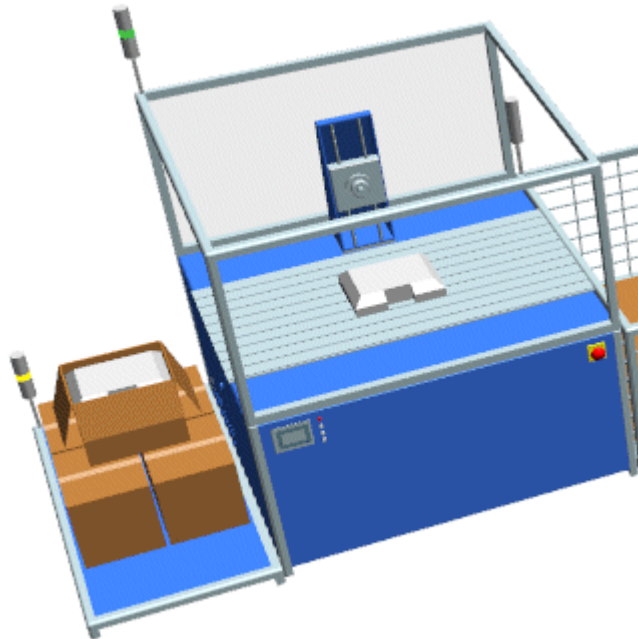
When you select to show **vertically arranged state graphics** for a machine for which you imported or created a graphic, you will have to adjust the position settings of the state graphics so that *Plant Simulation* show them correctly without overlapping a part of the machine.

By default, vertical states are shown with a signal lamp on a gray signal post positioned at the left top rear corner of the standard graphics. This way you don't have to visualize the states to be able to move them around without the dialog **Edit 3D Properties**.

To activate **vertically arranged state graphics**, open the object in **3D**, click the right mouse button into the background of the 3D window, and select **Edit 3D Properties**.



Our default graphic of the *Station* with vertically arranged state graphics on a signal post looks like this:



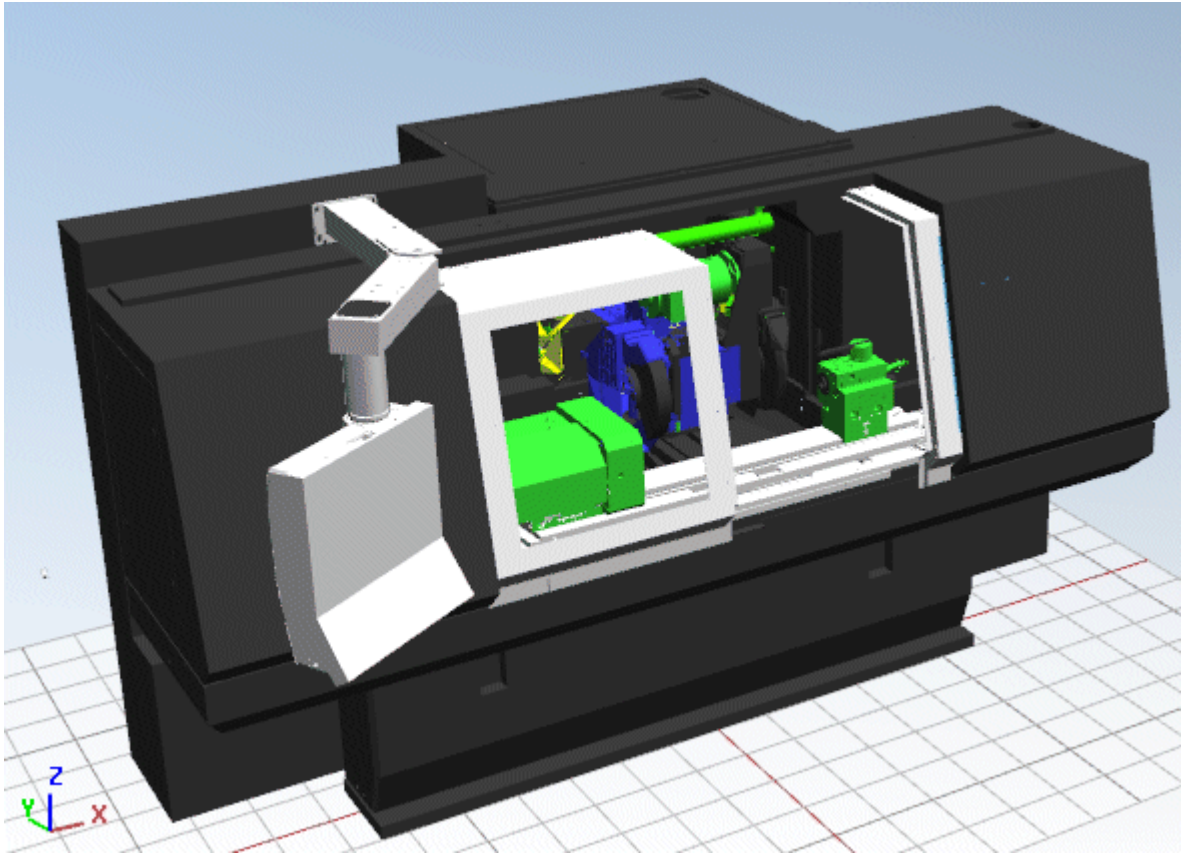
Back to [Show Object States](#)

Show States at the Front of the Machine

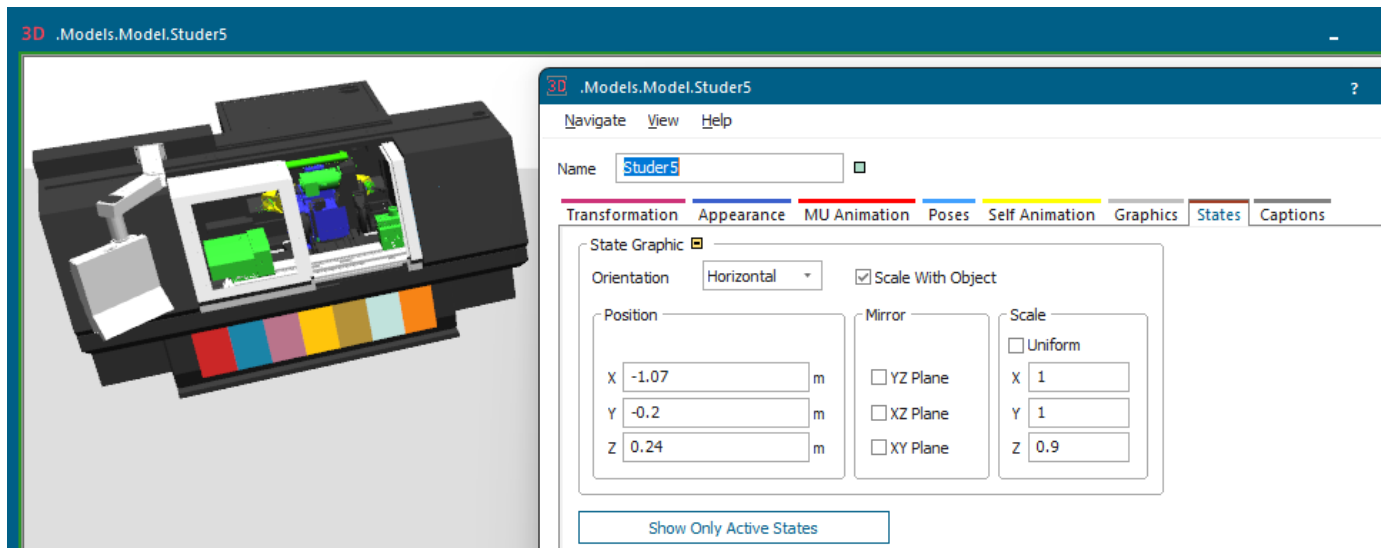
Plant Simulation show the states as **horizontally arranged cubes** at the back of the graphic of the machine by default.

Depending on the look of the graphic of the machine this may not be practical as the graphic of the machine might hide the state graphics as such. To fix this, we'll demonstrate how to show the state graphics at the front of the machine.

Our sample graphic without state graphics looks like this:



To activate **horizontally arranged cubes**, open the object in 3D, click the right mouse button into the background of the 3D window, and select **Edit 3D Properties**.



- To show the state graphics, click **Show All Possible States**. Rotate the machine to show the state graphics. You'll notice that the default setting is not helpful as all state graphics are hidden by the machine when we look at it from the front and as some of them disappear inside of the machine.
- To fix that, let's move the horizontal states to the front of the machine. You can either
 - **Move the state graphics with the arrow keys.** Select the state graphics bar. Press the **Up** or **Down arrow** to move it in the **Y** direction. Press the **Left** or **Right arrow** to move it in the **X** direction. Hold down **Ctrl** and press the **Up** or **Down arrow** to move it in the **Z** direction. Or you can
 - **Move the state graphics with the mouse.** Select the state graphics bar. Drag the mouse left and right to move it in the **Y** direction. Drag the mouse back and forth to move it in the **X** direction. Hold down the **Ctrl** key and drag the mouse up and down to move it in the **Z** direction.

You'll notice that moving objects with the arrow keys and/or the mouse might not be accurate enough. Instead, you can also enter precise values into the dialog **Edit 3D Properties**. If you want to, you can also combine the three methods by first moving the object with the arrow keys, or the mouse, to roughly position the object where you want it to be and by then entering precise values into the dialog.

- Click the machine with the right mouse button and select **Show Graphic Structure**. Then select **Edit 3D Properties** in the window **Show Graphic Structure**.
- Change the position on the tab **Transformation** so that **X** reads **-1.07**, so that **Y** reads **-0.2**, and that **Z** reads **0.25**.
- Click **Apply** to check if the result looks like we expected it to be. The state graphics bar moved to the front of the machine, but it is too high. To fix that, we scale it in the **Z** direction. Clear the check box **Uniform** and enter **0.9** into the text box **Z**. Now it looks good.

Compare States of the Material Flow Objects

Back to Show Object States